Development of a multimodal emotion recognition system using attention bottleneck mechanism

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Abstract

This diploma project presents the development and evaluation of a multimodal emotion recognition system using an attention bottleneck mechanism. The system is designed for the CMU-MOSEI six-emotion multi-label classification task and processes three modalities: text, audio, and visual information. The target emotion categories are happiness, sadness, anger, surprise, disgust, and fear. The proposed model combines modality-specific encoders with a bottleneck attention fusion module. Textual information is represented using a frozen BERT-base-uncased encoder with averaged hidden-layer representations. Acoustic information is represented using COVAREP features, while visual information is represented using OpenFace facial actionunitfeatures.Themodalityrepresentationsareprojectedintoasharedhiddenspace and fused through learnable bottleneck tokens, which provide an efficient communication channel between modalities without full pairwise cross-modal attention. The model was trained and evaluated on the CMU-MOSEI dataset using macro averaged F1 score as the main evaluation metric due to class imbalance. Additional valuation was performed using average weighted accuracy and per-class F1 scores. The proposed bottleneck fusion model outperformed the late fusion baseline, achieving an Avg. F1 of 0.4854 with the default threshold and 0.4942 after per-class threshold tuning. The results show that bottleneck-based fusion improves multimodal interaction and provides better recognition of minority emotion classes, especially fear and surprise. In addition to the experimental model evaluation, the project includes a functional system prototype. A FastAPI backend was implemented to process raw video input by extracting audio, visual, and textual features using ffmpeg, opensmile, py-feat, faster- whisper, and BERT tokenization. The trained model is then used to return emotion probabilities through a REST API. The system demonstrates the practical applicability of the proposed approach as a research prototype for multimodal emotion recognition.

Keywords: multimodal emotion recognition, CMU-MOSEI, attention bottleneck mechanism, BERT, COVAREP, OpenFace, macro-F1, multi-label classification.

Dedication and acknowledgements

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Author's declaration

I declare that the work in this diploma work was carried out in accordance with the requirements of the University's Regulations and that it has not been submitted for any other academic award. Except where indicated by specific references in the text, the work is the candidate's own work. Work done in collaboration with, or with the assistance of, others, is indicated as such. Any views expressed in the dissertation are those of the author.

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Definitions

Following terms are used in this work:

Multimodal Emotion Recognition	The task of identifying human emotions using multiple data modalities such as text, audio, and video.
Attention Bottleneck Mechanism	A fusion mechanism that enables efficient interaction between different modalities through a limited set of shared bottleneck tokens.
Modality	A specific type of input data used in the system, for example textual, acoustic, or visual information.
Fusion	The process of combining information from multiple modalities into a unified representation for emotion prediction.
Transformer	A neural network architecture based on self-attention mechanisms for modeling sequential data and contextual relationships.
Self-Attention	A mechanism that allows a model to determine the importance of different elements within the same sequence.
Bottleneck Tokens	Learnable latent representations used to exchange compressed information between modalities in attention bottleneck models.
Emotion Classification	The process of assigning an input sample to one of several predefined emotion categories.
Feature Extraction	The process of transforming raw input data into informative numerical representations suitable for machine learning models.

Text Modality	Linguistic information represented as textual features or embeddings extracted from speech transcripts.
Audio Modality	Acoustic information such as tone, pitch, energy, and speech characteristics extracted from audio signals.
Video Modality	Visual information including facial expressions, gestures, and movements extracted from video frames.
BERT	A pretrained transformer-based language model used for extracting contextual textual embeddings.
COVAREP	A speech analysis toolkit used for extracting acoustic features from audio data.
OpenFace	An open-source toolkit for facial behavior analysis and extraction of facial action units.
CMU-MOSEI	A large-scale multimodal dataset containing video clips annotated with sentiment and emotion labels.
Weighted F1-score	An evaluation metric that measures classification performance while considering class imbalance.
Cross-Entropy Loss	A loss function commonly used in classification tasks to measure the difference between predicted and true labels.

Designations and Abbreviations

Following designations and abbreviations are used in this work:

AI	Artificial Intelligence
MER	Multimodal Emotion Recognition
HCI	Human–Computer Interaction
CMU-MOSEI	Carnegie Mellon University Multimodal Opinion Sentiment and Emotion Intensity Dataset
MELD	Multimodal EmotionLines Dataset
BERT	Bidirectional Encoder Representations from Transformers
CNN	Convolutional Neural Network
BiLSTM	Bidirectional Long Short-Term Memory
Transformer	Attention-based neural network architecture for sequence modeling
MulT	Multimodal Transformer
DBA	Domain-Separated Bottleneck Attention
XMBT	Extended Multimodal Bottleneck Transformer
FFN	Feed-Forward Network
ReLU	Rectified Linear Unit
BCEWithLogitsLoss	Binary Cross-Entropy Loss with Logits
F1-score	Harmonic mean of precision and recall
Macro-F1	Average F1-score calculated equally across all classes
Avg. WA	Average Weighted Accuracy

TP	True Positive
TN	True Negative
GPU	Graphics Processing Unit
API	Application Programming Interface
WS	WebSocket
COVAREP	Collaborative Voice Analysis Repository for Speech Technologies
OpenFace	Open-source facial behavior analysis toolkit
openSMILE	Open-source Speech and Music Interpretation by Large-space Extraction toolkit
AU	Facial Action Unit
PyTorch	Open-source deep learning framework

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Chapter 1

Introduction

Emotion recognition has become an important research direction in artificial intelligence because modern digital communication increasingly relies on text, audio, and video. People express emotions not only through the meaning of words, but also through voice tone, facial expressions, and other non-verbal signals. As communication moves into online platforms, video conferencing, educational environments, and intelligent assistants, automatic understanding of human affect becomes more relevant for both research and practice.

In affective computing, emotion recognition aims to identify human emotional states from observable signals. Such systems may be useful in human–computer interaction, digital education, mental health monitoring, social media analysis, and assistive technologies. Kalateh et al. describe multimodal emotion recognition as a field that combines different sources of human expression in order to improve the reliability of emotion analysis [4]. This is especially important because human emotions are complex and cannot always be correctly interpreted from a single source of information.

Unimodal systems rely on only one modality, such as text, speech, or facial expression, which has clear limitations. Text may show semantic meaning, but it does not fully describe how the speaker said something. Audio can capture vocal cues such as pitch and rhythm, but may not contain enough semantic information. Visual features represent facial expressions and head movements, but they may be unreliable when video quality is low. For this reason, multimodal emotion recognition combines textual, acoustic, and visual information to obtain a more complete representation of human emotion [7].

The CMU-MOSEI dataset is one of the most widely used benchmarks for multimodal sentiment and emotion analysis. Zadeh et al. introduced it as a large-scale dataset collected from online videos, containing text, audio, and visual modalities [20]. CMU-MOSEI includes six emotion categories: happiness, sadness, anger, fear, disgust, and surprise. A major challenge is class imbalance, as happiness accounts for over 53% of samples, while fear and surprise are rare. Nguyen et al. emphasize that imbalanced datasets require special attention to minority classes and appropriate evaluation metrics [11]. For this reason, this project uses macro-F1 as the main metric, which gives equal importance to all emotion classes.

Another key challenge is multimodal fusion. Transformer-based models such as the Multimodal Transformer by Tsai et al. introduced cross-modal attention for unaligned sequences and demonstrated the importance of learning interactions between modalities [17]. However, full cross-modal attention can be computationally expensive. Attention bottleneck mechanisms address this limitation. Nagrani et al. proposed the Multimodal Bottleneck Transformer, where a small set of bottleneck tokens controls information exchange between modalities instead of allowing all tokens to attend to each other directly [10]. Later works adapted this idea to emotion recognition. He et al. proposed a domain-separated bottleneck attention fusion framework [2], and Nguyen et al. extended bottleneck transformers to address class imbalance through adaptive loss design [11].

Research Aim

The aim of this diploma project is to develop and evaluate a multimodal emotion recognition system based on an attention bottleneck mechanism for six-emotion classification using textual, acoustic, and visual modalities from the CMU-MOSEI dataset.

Research Objectives

To achieve this aim, the following objectives are defined:

- to study existing approaches to multimodal emotion recognition and attention-based fusion methods;
- to analyze the CMU-MOSEI dataset and prepare multimodal features for training;

- to design and implement a multimodal architecture using an attention bottleneck mechanism;
- to train the model under class imbalance conditions using weighted loss and regularization techniques;
- to evaluate the system using macro-F1, weighted accuracy, and per-class metrics;
- to compare the proposed bottleneck fusion approach with a simpler fusion baseline.

The research is guided by several main questions. First, how can textual, acoustic, and visual features be effectively combined for emotion recognition on CMU-MOSEI? Second, whether an attention bottleneck mechanism can improve multimodal fusion compared to a simpler late fusion baseline. Third, how class imbalance affects recognition of rare emotions such as fear and surprise. Fourth, which evaluation metrics provide the most meaningful assessment of model performance in an imbalanced multimodal setting.

The scope of this project is limited to prototype-level development and experimental evaluation. The system uses precomputed multimodal features instead of raw audio and video processing, which allows the research to focus on multimodal fusion rather than low-level feature extraction. The system is not intended as a production-ready real-time application.

The rest of this report is organized as follows. Chapter 2 reviews existing research on multimodal emotion recognition, CMU-MOSEI, feature extraction, fusion methods, and bottleneck attention mechanisms. Chapter 3 describes the research methodology, dataset preparation, preprocessing, evaluation metrics, and experimental setup. Chapter 4 presents the system design and implementation. Chapter 5 reports the experimental results. Chapter 6 discusses the findings, limitations, and future work. Chapter 7 summarizes the main conclusions.

Chapter 2

Literature Review

2.1 Emotion Recognition in Human Communication

Emotions are a fundamental part of how people communicate in nowadays world. When someone speaks, they do not only use words to express their thoughts — they rely on the tone of their voice, their facial expressions, and their body language to convey how they feel. For a long time, computers were simply able to understand the content of speech, not the way in which it was spoken or the speaker's facial expressions. The field of emotion recognition research aims to alter this by allowing machines to recognize and analyse human emotional states from observable information.

Kalateh et al. conducted a comprehensive systematic review of multimodal emotion recognition and described it as a field that "combines different types of human signals in order to improve the reliability of emotion analysis" [4]. Their review covers a wide range of applications where emotion recognition is already being used or studied: human-computer interaction, healthcare, education, social robotics, gaming, and intelligent assistants. What stands out in their findings is that none of these domains can rely on a single signal alone — in healthcare, for example, a patient might say they feel fine while their voice and face suggest otherwise. This gap between what is said and what is actually felt is exactly why automatic emotion recognition is both difficult and important.

The field did not start with the sophisticated systems we see today. Koromilas and Giannakopoulos traced the development of multimodal emotion recognition from its earliest stages, showing that the first systems relied almost entirely on handcrafted features and classical machine learning classifiers like Support Vector Machines [7]. These early methods worked decently in controlled laboratory environments, but they had trouble with real-world data, including noisy speech, partially visible faces, and subtle or ambiguous emotional expressions. This was fundamentally changed as the transition toward deep learning; recurrent neural networks (RNN), convolutional networks, and eventually transformer-based models. All this made it possible to learn complex patters straight from data without needing to manually describe what an emotional signal looks like.

One of the core reasons multimodal approaches outperform unimodal ones is explained clearly by Kalateh et al., who note that using several modalities together allows the system to become more robust, because one modality can compensate for the weakness of another [4]. A classic example is sarcasm — a sentence that sounds positive in text might carry a completely different meaning when combined with a flat or negative vocal tone. Similarly, a spoken phrase might be unclear without seeing the speaker's face. Neither text, audio or video alone tells the full story. Koromilas and Giannakopoulos further support this by pointing out that "the two key challenges in multimodal learning are learning complex interactions within and across modalities, and building models that are robust when one modality is missing or noisy at test time" [7].

This project sits directly within this context. The system developed there by using an attention bottleneck fusion technique to deal with three modalities: text, audio, and visual. The purpose is to learn how various signals interact and what occurs when the model must handle uncommon emotions that appear infrequently in training data. This goal goes far beyond merely creating a system that accurately identifies the most prevalent emotion. The challenges highlighted by Kalateh et al. directly shaped the design of this project. Specifically, modality heterogeneity, class imbalance, and the necessity for effective cross-modal interaction motivated the architectural and training choices detailed in subsequent chapters, a consensus further supported by Koromilas and Giannakopoulos[7].

2.2 The CMU-MOSEI Dataset

Datasets are important in multimodal emotion recognition because they define what kind of emotional signals a model can learn from. For this task, the dataset should include text, audio, and visual information, since emotion is usually expressed through language, voice, and facial behavior together. Zadeh et al. introduced CMU-MOSEI as a large-scale collection for multimodal language analysis collected from online videos [20]. The authors describe it as a dataset based on natural monologue-style recordings, where speakers express opinions in real-world conditions. This makes it useful for realistic emotion recognition research, but also more difficult than controlled laboratory datasets.

In the original paper, Zadeh et al. explain that the dataset contains three main modalities: language, acoustic signals, and visual signals [20]. Each modality carries a different type of affective information. Text provides the semantic meaning of an utterance. Audio reflects how the speaker says it through prosody, rhythm, and vocal intensity. Visual information describes facial behavior and other non-verbal cues. This multimodal structure makes CMU-MOSEI suitable for studying cross-modal learning, because the model has to combine different sources of information instead of relying on only one channel.

Main study collection is also important because it supports both sentiment analysis and emotion recognition. Sentiment analysis usually focuses on polarity, such as positive or negative meaning, while emotion recognition is more detailed and focuses on separate emotion categories. Mora Melanchthon describes CMU-MOSEI emotion recognition as a categorical task where each emotion can be treated as present or absent [9]. This means that the task is not the same as simple single-label classification. One utterance may contain several emotional cues, and each emotion must be considered separately.

A major difficulty of CMU-MOSEI is the uneven distribution of emotion labels. Mora Melanchthon notes that happiness is the most frequent emotion in the dataset, while fear and surprise are among the rarest categories [9]. This imbalance affects how models learn. A model may recognize frequent emotions more easily, but fail on rare classes. Because of this, overall accuracy can be misleading. A system may look effective if it predicts common emotions correctly, even when it performs poorly on minority emotions.

Mora Melanchthon also shows that feature-level performance on dataset differs across modalities [9]. Text, acoustic, and visual features do not contribute equally to every emotion, and feature selection or transformation can affect model performance. This shows that multimodal emotion recognition depends not only on the fusion method, but also on how well each modality is represented before fusion. Overall, Zadeh et al. provide the foundation for CMU-MOSEI and its multimodal structure, while Mora Melanchthon highlights its emotion recognition setting, class imbalance, and modality-specific feature challenges [9, 20].

2.3 Feature Extraction for Text, Audio, and Video Modalities

Before a multimodal system can start learning, raw data must be converted into numerical representations that can be processed by a neural network. A video file itself is only a combination of visual frames and sound signals, and these signals need to be transformed into structured feature representations. This stage is known as feature extraction. In this project, three separate feature pipelines are used: one for text, one for audio, and one for visual data. Each pipeline is based on established tools and methods used in multimodal emotion recognition literature.

For the text modality, recent studies often use pre-trained language models instead of simple static word embeddings. Makhmudov et al. explain that using a pre-trained BERT-based model is useful because it has already learned rich linguistic representations from a large text corpus [8]. BERT, which stands for Bidirectional Encoder Representations from Transformers, processes text bidirectionally and therefore captures how each word is related to its surrounding context. This is important for emotion recognition because the emotional meaning of a word or phrase often depends on the full sentence. For example, the word “fine” may express a neutral meaning in one context but may also carry negative or sarcastic meaning in another context. In this project, BERT-base-uncased is used as a frozen feature extractor, and the final text representation is obtained by averaging the outputs of the last four hidden layers. A similar use of BERT-based textual representation is also found in video emotion recognition research by Xia et al. [18].

For the audio modality, this project uses COVAREP acoustic features. COVAREP is a collaborative voice analysis toolkit that extracts low-level acoustic descriptors from speech signals. It produces a 74-dimensional feature vector for each time frame and captures properties such as fundamental frequency, Mel-frequency cepstral coefficients, spectral envelope, glottal source parameters, and voiced–unvoiced segmentation indicators. These acoustic descriptors represent how a person speaks, including pitch, rhythm, energy, and voice quality. Such characteristics are important for emotion recognition because emotional states are often reflected in vocal behavior. For example, fear may be associated with changes in pitch and energy, while sadness may be expressed through slower and lower-energy speech. Mora Melanchthon discusses the use of acoustic features in CMU-MOSEI-based emotion recognition and shows that such features provide a compact and interpretable representation of emotional speech signals [9]. COVAREP-based acoustic features are also commonly used in multimodal transformer-based studies on CMU-MOSEI [17].

For the visual modality, this project uses facial behavior features extracted with OpenFace. OpenFace is a toolkit that detects facial landmarks, head pose, eye gaze, and facial action units from video frames [9]. Facial action units describe the movement of specific facial muscle groups. For example, some action units correspond to cheek movement, lip corner movement, or brow movement. In this project, the visual representation consists of 35 values per frame, including action unit intensity scores and binary presence indicators. These features are not raw pixel values; they are compact numerical descriptors of facial behavior at each moment. This makes them easier to process and more interpretable than raw image frames, especially in datasets such as CMU-MOSEI where video quality and recording conditions may vary.

The use of handcrafted features such as COVAREP and OpenFace is a deliberate trade-off in this project. More recent models, such as the MER-SEM-MBT model proposed by Xia et al., use stronger pretrained neural encoders, including VGGish for audio and ResNet18 for visual features [18]. These encoders can produce richer representations and may achieve stronger benchmark results. However, they also require more computational resources and are less interpretable than compact handcrafted descriptors. Therefore, this project uses standard CMU-MOSEI feature representations in order to keep the system computationally feasible and comparable with widely used multimodal baselines such as MulT [17].

2.4 Multimodal Fusion Approach

When features are extracted from each modality, the next question is how to combine them. In general, this step is calles multimodal fusion, and it isplays a crucial role in the design of multmodal system. The modality fusion strategy determines whether the model can effectively capture cross-modal relationships between rext, audio and video. Over the years,researchers have proposed many different fusion strategies, varying from straightforward to complex architectures. Therefore, undestanding their trade-offs is very important for the frameworks.

The simplest strategies are early fusion and late fusion. In early fusion, features from all input streams are concatenated into a single vector before being passed to the model.

This is easy to implement, but it assumes that all modalities are equally important at every stage of processing. That is why one madality may be noisier or weaker than the others. In late fusion, each modality is processed by a completely separate model, and only the final predictions are combined. It allows each modality to be processed independently, but it reduces the capacity to learn fine-grained cross-modal feature interactions. As Nagrani et al. describe, late fusion is a useful baseline. However, full early fusion, where every token can attend to every other token across modalities, is computationally expensive and not always more accurate. This happens because much of the cross-modal attention is spent on redundant or irrelevant information [10].

The approach that changed the direction of the field for multimodal language analysis was the Multimodal Transformer, known as MulT, introduced by Tsai et al. [17]. Instead of simply concatenating features or averaging predictions, MulT introduced directional cross-modal attention, where one modality attends to another. For example, the language modality can attend to the audio sequence, asking at each word: which audio frames are most relevant to what is being said here? This property is particularly useful for CMU-MOSEI, where the modalities are not perfectly aligned in time. For instance, a facial expression may appear a fraction of a second before the corresponding word is spoken. MulT handles this naturally because cross-modal attention does not require alignment; it learns the correspondence between sequences directly from data. Tsai et al. showed that cross-modal attention consistently outperforms early and late fusion approaches on CMU-MOSEI. As a result, MulT has become a widely used benchmark in later studies.

Nevertheless, MulT has a limitation. When three modalities are used with bidirectional attention between every pair, the number of cross-modal attention operations increases rapidly. This leads to a situation where a considerable amount of attention is spent on information that is not always relevant for the final emotion prediction. Jia et al. also note that the audio modality often contributes less than text or vision in many multimodal models. They argue that more efficient fusion strategies can help balance the contribution of each modality without significantly reducing overall performance [3]. This observation is relevant to this project, where COVAREP audio features are already weaker than BERT text representations by nature, and the fusion mechanism needs to handle this imbalance thoughtfully.

Recent work has focused on combining the advantages of cross-modal attention with more controlled information flow. Xia et al. introduced a model that adds a semantic enhancement step before fusion. It uses the text modality to guide how audio and visual representations are refined [18]. Their method is based on the assumption that text provides the most reliable high-level semantic information. By making audio and visual representations more text-aware before fusion, the model can learn better joint representations.This type of modality-guided fusion represents a more structured way of modeling cross-modal interactions. Instead of allowing all modalities to influence each other equally, some models explicitly assign a guiding role to the most informative modality.

All of these approaches share a common limitation. They allow full pairwise attention between all tokens across modalities, which scales quadratically with sequence length. As a result, the model exchanges a large amount of redundant information. The next section introduces a fundamentally different idea that this project builds on directly. The attention bottleneck restricts cross-modal information flow by passing it through a small set of shared latent tokens.

2.5 Attention Bottleneck Mechanism

More controlled way for modality of exchange information demonstrated by attention bottleneck mechanism. In emotion recognition, this is especially useful because text, audio, and visual signals do not contribute equally at every moment. Some words may be emotionally neutral, some audio frames may contain noise, and some facial movements may not express a clear body language. Full cross-modal attention can capture rich interactions, but it may also become computationally heavy and pass redundant information between modalities. Nagrani et al. address this problem through the Multimodal Bottleneck Transformer, where a small group of fusion tokens works as an intermediate communication channel between modality-specific streams [10]. Similar concerns are also discussed by He et al., who use bottleneck attention in a domain-separated fusion framework to control interaction between shared and modality-specific representations [2]. Nguyen et al. further promote this direction by extending bottleneck-based fusion to multimodal emotion recognition under class imbalance conditions [11]. Together, these studies show that bottleneck attention is not only a way to reduce computation. They act as a method for making cross-modal communication more selective and meaningful.

Nagrani et al. introduced this idea in the Multimodal Bottleneck Transformer (MBT) [10]. Instead of allowing focused dedication between all modality tokens, MBT uses a small number of fusion bottleneck tokens as an intermediate communication channel. Each modality sends information to these tokens, and the bottleneck tokens then share the compressed information back to the modality-specific streams. Their study shows that bottleneck-based fusion can reduce computational cost while still preserving important cross-modal interaction. This makes the mechanism useful for multimodal tasks where full pairwise attention may be unnecessary or inefficient.

The main strength of bottleneck tokens is their selectivity. Since the number of bottleneck tokens is limited, the model cannot simply transfer all information from one modality to another. It has to compress and prioritize the most important signals. Because emotional evidence is often localized to brief vocal or facial expressions, bottleneck attention provides a dual benefit: reducing computational costs and guiding the network toward sparse, meaningful multimodal features [10].

Later studies adapted bottleneck-based fusion more directly to multimodal emotion recognition. He et al. proposed the Domain-Separated Bottleneck Attention Fusion framework, where modality representations are separated into private and invariant domains [2]. The private domain preserves modality-specific information, while the invariant domain captures shared emotional information across modalities. This is useful because text, audio, and visual signals do not always express emotion in the same way. By applying bottleneck attention between these separated representations, the model can exchange information in a more structured way instead of simply mixing all features together [2].

Nguyen et al. also extended the bottleneck transformer approach in the Extended Multimodal Bottleneck Transformer (XMBT) [11]. Their work is especially relevant to CMU-MOSEI emotion recognition because it considers both multimodal fusion and class imbalance. XMBT uses modality-specific branches together with shared bottleneck tokens and adds auxiliary learning signals for individual modalities. The authors also introduce Dynamic Restrained Adaptive loss to enhance learning for underrepresented emotions. This showcase that bottleneck fusion integrates with targeted training strategies to mitigate modality dominance and class imbalance [11].

Overall, research points to computational and representational benefits of attention bottleneck approaches. They recommend the model to focus on clear, informative cross-modal signals and prevent unneeded attention exchanges. Nevertheless, bottleneck attention by itself is not a whole solution. The training strategy, class imbalance management, and the quality of modality-specific elements continue to determine its efficacy. Bottleneck fusion is therefore combined with domain separation, secondary losses, and adaptive loss functions in order in modern studies [2, 11]. These findings provide the theoretical foundation for this project's primary fusion approach that acts as an attention bottleneck mechanism.

2.6 Gaps in Existing Research and Project Justification

The reviewed literature shows that multimodal emotion recognition has improved significantly, but several problems remain open. One of the main difficulties is the recognition of rare emotions in CMU-MOSEI. Emotions such as fear and surprise appear much less often than happiness, so models can become biased toward frequent categories. Nguyen et al. discuss this problem and propose adaptive loss design to improve learning for underrepresented emotions [11]. He et al. also show that minority emotions are harder to classify because of class imbalance and overlapping emotional patterns [2]. For this reason, this project does not rely only on general accuracy. It considers weighted loss, threshold tuning, macro-F1, and per-class F1 analysis.

Another important gap concerns how different modalities are represented before fusion. Text, audio, and visual features have different structures and dimensions, so they uncapable to fuse directly without transformation. Encoders are needed to convert each modality into a numerical representation that the model can process, and to map all modalities into a shared hidden dimension. Some recent studies use more complex feature extraction pipelines at this stage. For example, Xia et al. use VGGish for audio and ResNet18 for visual information to obtain richer representations [18]. However, such approaches require more computational resources. This project utilizes standard CMU-MOSEI features. Specifically, modality-specific encoders project BERT-based text, COVAREP acoustic, and OpenFace-style visual features into a unified representation space prior to bottleneck fusion.

A further limitation is that many studies make it difficult to understand which part of the model actually improves performance. In some papers, several changes are added at the same time: stronger feature extractors, new fusion blocks, auxiliary losses, or adaptive training strategies. As a result, it is not always clear whether the improvement comes from the fusion mechanism or from another component. Xia et al. show the value of semantic enhancement, while Nguyen et al. combine bottleneck fusion with auxiliary losses and adaptive loss design [11, 18]. This project addresses the same issue by comparing model variants under one experimental setting and analyzing the effect of different design choices.

Furthermore, a practical gap persists between benchmark models and deployable, production-ready systems. While literature extensively reports performance metrics on curated datasets, discussions regarding the integration of these models into operational applications remain limited. Kalateh et al. note that computational complexity, real-time performance, and deployment remain important challenges in multimodal emotion recognition [4]. This is relevant because an emotion recognition model should not only work in an isolated experiment, but also be understandable and extendable as part of a software system. Consequently, this project evaluates both model training and the deployment of the trained architecture within a prototype system.

In conclusion, these gaps justify the direction of this work. The project focuses on attention bottleneck fusion because it offers a controlled way to exchange information between text, audio, and visual modalities. Furthermore, this study addresses class imbalance, modality representation, and systematic evaluation. The objective extends beyond comparative score analysis to evaluate whether a compact, bottleneck-based fusion strategy can effectively support a realistic multimodal emotion recognition system utilizing standard CMU-MOSEI features.

2.7 Comparative Analysis of Related Works

Table 2.1 summarises the main related works reviewed in this chapter. The comparison focuses on the model type, used modalities, dataset, reported Avg. F1 where available, and the design idea that makes each study relevant to this project.

Table 2.1: Comparative analysis of related works in multimodal emotion recognition.

Study	Model Type	Modality	Dataset	Avg. F1	Key Notes
Tsai et al. (2019) [17]	Multimodal Transformer (MulT)	Text + Audio + Vision	CMU-MOSEI	0.475	Uses directional cross-modal attention for unaligned sequences. This work is important as a strong reference model for CMU-MOSEI.
Nagrani et al. (2021) [10]	Multimodal Bottleneck Transformer (MBT)	Audio + Vision	AudioSet, VGGSound	–	Introduces bottleneck tokens for multimodal fusion. The idea reduces full pairwise attention by passing information through shared latent tokens.
Xia et al. (2022) [18]	MER-SEM-MBT	Text + Audio + Vision	CMU-MOSEI	0.509	Combines semantic enhancement with bottleneck fusion. The model uses BERT, VGGish, and ResNet18, which gives stronger audio and visual representations.
He et al. (2024) [2]	Domain-Separated Bottleneck Attention	Text + Audio + Vision	CMU-MOSEI	–	Separates private and invariant modality representations. The study also supports the use of bottleneck attention for structured cross-modal exchange.
Nguyen et al. (2025) [11]	XMBT + DRA loss	Text + Audio + Vision	CMU-MOSEI	0.496	Extends bottleneck fusion to multimodal emotion recognition and adds adaptive loss for class imbalance. The work is especially relevant for rare emotions such as fear and surprise.
Sanjyal (2026) [14]	Causal-Diffusion Bridge	Text + Audio + Vision	CMU-MOSEI subset	0.214	Uses compact acoustic and visual features with a smaller aligned subset. The result shows that feature quality and data volume can strongly affect final scores.
Proposed work	**BERT + Conv1D + Bottleneck Attention**	**Text + Audio + Vision**	**CMU-MOSEI**	**0.485**	Uses frozen BERT, COVAREP, OpenFace, 16 bottleneck tokens, auxiliary losses, and modality dropout. The model focuses on compact feature representations and rare emotion recognition.

Chapter 3

Methodology

3.1 Research Design

This work is applied in nature because it focuses on developing and evaluating a functional multimodal emotion recognition system. It does not attempt to propose a new theory of human emotion. Instead, it addresses a practical machine learning task: combining text, audio, and visual information to classify six emotion categories in an imbalanced dataset. The main outcome is an operational neural network and a supporting processing pipeline that can transform multimodal inputs into emotion predictions.

The study follows a design-oriented research logic. The process begins with problem analysis through the literature review, where existing approaches, limitations, and evaluation practices are examined. Based on this analysis, the architecture is designed and then implemented. After that, the baseline and proposed models are trained under the same experimental conditions. The final stage is evaluation using quantitative metrics. This sequence is suitable for deep learning research, where architectures are commonly tested through controlled experiments on established datasets. Koromilas and Giannakopoulos also describe deep multimodal emotion recognition as a field where model development and experimental comparison are central parts of the research process [7].

The work has two connected directions. The first direction is experimental and focuses on model performance. Two baselines and the proposed bottleneck attention architecture are trained on CMU-MOSEI and compared under the same validation and test protocol. The first baseline uses late fusion with a BiLSTM visual encoder and no cross-modal interaction. The second baseline adds cross-modal attention between all three streams, following the MulT design. This setup answers the main research question about whether structured bottleneck communication improves recognition beyond simpler fusion strategies. The second direction is practical and concerns prototype integration. The trained model is prepared for use inside a processing pipeline that connects to the MultiMOOD web interface. In this way, the study is not limited to a notebook experiment, but also considers how the developed solution works in an applied software setting.

The evaluation is quantitative. Interviews, surveys, and subjective user assessment are not included because the research questions focus on measurable model performance. The evidence is obtained from macro-F1, weighted accuracy, and per-class F1 scores computed on validation and test sets. These metrics make it possible to compare different variants and examine whether the proposed architecture improves recognition, especially for underrepresented emotion categories.

3.2 Quantitative Experimental Approach to Emotion Recognition

This study uses a quantitative experimental approach because its main objective is to measure the performance of machine learning models. In emotion recognition research, model quality is usually evaluated through numerical metrics computed on validation and test data. In this work, the main measures are macro-averaged F1 score, average weighted accuracy, and per-class F1 scores for the six emotion categories. These values allow the proposed bottleneck-based architecture to be compared with two baselines under identical conditions. Ramyasree et al. use a similar evaluation logic in their multimodal bottleneck transformer study, where accuracy and F1-score are reported as the main indicators of performance [13].

The quantitative approach is also important because CMU-MOSEI is imbalanced. A general accuracy score may not show whether rare emotions are recognized well. For this reason, macro-F1 and per-class F1 are used to examine results across all categories, not only the dominant ones. This is especially relevant for difficult classes such as fear and surprise.

The same logic is applied to the ablation study. Several configurations are tested under one controlled setup. Each configuration changes a specific part of the architecture or training strategy, such as BERT freezing, the number of averaged BERT layers, auxiliary losses, modality dropout, or threshold tuning. Macro-F1, weighted accuracy, and per-class F1 are then compared to determine the effect of each design choice. This makes the analysis more systematic and helps identify which components contribute most to the final result.

3.3 Justification of the Experimental Evaluation Approach

The experimental evaluation approach was selected because it matches the purpose of this work. The central question is whether the attention bottleneck mechanism improves multimodal emotion recognition compared with simpler fusion strategies. This type of question is best answered through controlled experiments, where all three architectures train and test on the same data under the same conditions. Interviews or subjective observations would not provide the main evidence in this case, because model quality must be assessed through measurable results on unseen samples.

Using a widely accepted dataset and standard metrics also makes the findings easier to compare with previous studies. Koromilas and Giannakopoulos describe deep multimodal emotion recognition as a field where architectures are often evaluated through experimental comparison and numerical results [7]. CMU-MOSEI is suitable for this purpose because it is widely used in multimodal emotion and sentiment recognition research. If a different dataset or unusual metrics were used, it would be harder to place the results in the context of existing work.

The selected metrics are connected to the properties of CMU-MOSEI. Since the data is imbalanced, plain accuracy may hide weak performance on rare emotions. Mora Melanchthon shows that macro-F1 is useful for CMU-MOSEI because it reflects recognition quality across separate categories more clearly than general accuracy [9]. Therefore, macro-F1 is used as the primary metric, while weighted accuracy is reported as a secondary measure. This strategy evaluates both overall behaviour and recognition of underrepresented classes.

3.4 Dataset and Computational Instruments

The main data source for this work is CMU-MOSEI. It provides the training, validation, and test samples used in the experimental part of the study. Zadeh et al. introduced CMU-MOSEI as a large-scale dataset for multimodal language analysis collected from online videos [20]. It contains more than 23,000 annotated video segments and includes text, audio, and visual information. This structure is suitable for the present task because the system is designed to recognize emotions from three complementary input streams.

CMU-MOSEI already contains structured multimodal features and emotion labels. Each sample can be represented through textual, acoustic, and visual streams. This allows modality-specific encoders to process each source separately before fusion. Such a format is important for the proposed bottleneck module, where different inputs are first projected into a shared representation space and then combined. Related studies also use acoustic and visual descriptors such as COVAREP and OpenFace-style features, which makes the dataset appropriate for comparison with previous work [9, 14].

The computational instruments include the software libraries and hardware environment used for preprocessing, training, evaluation, and prototype integration. PyTorch is used as the main deep learning framework for implementing both the baseline and proposed architectures. The HuggingFace Transformers library provides BERT-based text representations. Scikit-learn is used to compute macro-F1, weighted accuracy, and per-class F1. All training experiments are conducted on the Kaggle platform with a Tesla T4 GPU, which is sufficient for the selected model size and batch configuration.

Additional tools support the deployment-oriented preprocessing pipeline. The openSMILE toolkit extracts acoustic descriptors, Py-Feat detects facial action units, and faster-whisper transcribes speech. These tools allow the system to move beyond prepared dataset features and process raw video input in a prototype setting. Thus, the computational setup supports both experimental evaluation and practical implementation.

Figure 3.1 shows the tensor shape produced by each modality after preprocessing and projection. These shapes remain fixed across the experiments and define the input format expected by the bottleneck fusion model.

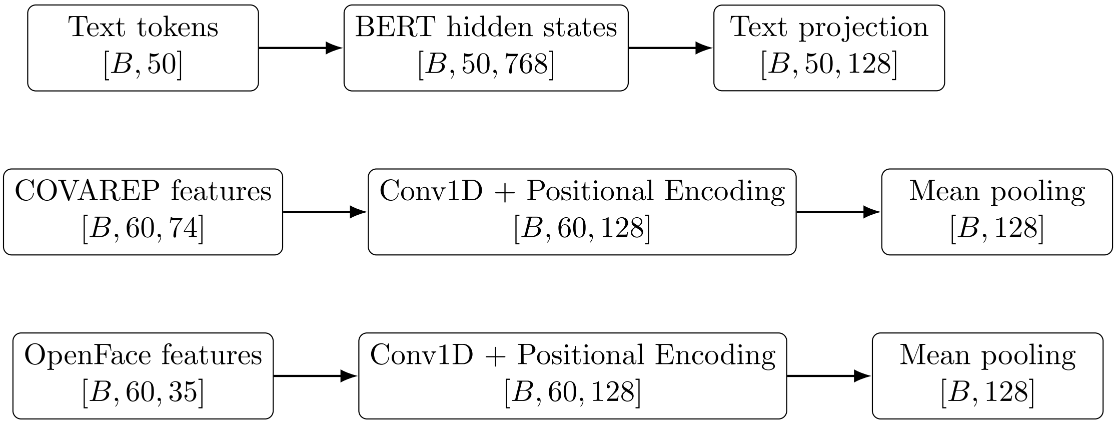

Figure 3.1: Feature tensor shapes for text, audio, and visual modalities.

3.5 Data Sampling and Class Imbalance in CMU-MOSEI

The experiments use the train, validation, and test split provided with CMU-MOSEI. A fixed split keeps the setting consistent and makes the results easier to compare with related studies. Before training, utterances with all-zero emotion labels are removed. Since this work focuses on six positive emotion categories, samples without any positive label are excluded from the supervised classification set. After filtering, the dataset contains 13,934 training samples, 1,569 validation samples, and 3,957 test samples. Nguyen et al. use a similar filtering strategy in their Extended Multimodal Bottleneck Transformer study on CMU-MOSEI [11].

The main sampling issue is class imbalance. The six emotion categories are not represented equally. Happiness is the dominant class and appears in approximately 62.7% of the training samples, while fear is the rarest category with only 9.6%. Surprise and disgust are also less frequent. This distribution affects learning because the classifier may recognize common emotions more easily and ignore minority classes. As a result, a model can show acceptable overall performance while still producing weak results for underrepresented categories. This is why macro-F1 and per-class F1 are important in this work.

Figure 3.2 shows the positive ratio for each emotion in the training set. The imbalance shown in the figure motivates several methodological decisions, including class-specific positive weights in the loss function and threshold tuning on the validation set. These choices are introduced later in this chapter and described in more detail in the system design and training sections.

$$pos\_weight_i = \frac{N_{negative,i}}{N_{positive,i}} \tag{3.1}$$

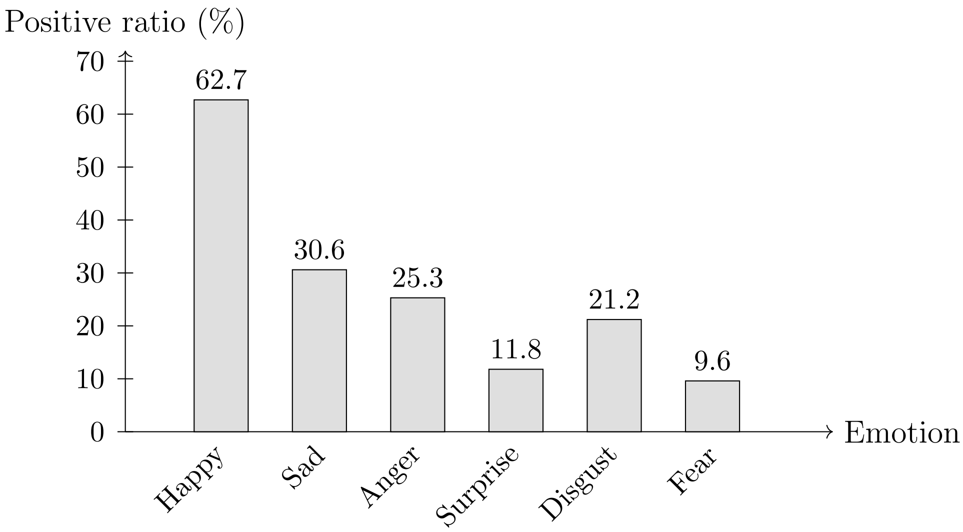

Figure 3.2: Class imbalance in the filtered CMU-MOSEI six-emotion dataset.

3.6 Multimodal Feature Preprocessing Pipeline

The preprocessing pipeline converts CMU-MOSEI multimodal features into a fixed tensor format required by the neural network. The original dataset introduced by Zadeh et al. contains text, audio, visual information, and emotion annotations collected from online video segments [20]. In this work, preprocessing does not create a new dataset from participants. It prepares the existing features for training, validation, testing, and later prototype inference.

For the text stream, the available word-level information is converted into sentence-level input and processed with a frozen BERT-base-uncased model. The final representation is obtained by averaging the last four hidden layers of BERT. This produces a tensor of shape $[B, 50, 768]$, where B is the batch size and 50 is the maximum sequence length. Before fusion, a linear projection maps this representation to the shared hidden dimension $[B, 50, 128]$.

$$H_{BERT} = \frac{H^{(9)} + H^{(10)} + H^{(11)} + H^{(12)}}{4} \tag{3.2}$$

For the audio stream, COVAREP descriptors are used. Each sequence contains 74-dimensional vectors over a fixed length of 60 time steps, producing a tensor of shape $[B, 60, 74]$. These values represent speech-related properties such as pitch, spectral characteristics, and voice quality. For the visual stream, OpenFace-style facial features are used. Each sequence contains 35-dimensional facial action unit descriptors over 60 frames, producing a tensor of shape $[B, 60, 35]$. Audio and visual sequences are padded when they are shorter than the required length, and all tensors are converted to 32-bit floating point format.

After preprocessing, the prepared samples are stored in a pickle file for efficient loading. Each item contains text, audio, and visual tensors, a six-dimensional binary emotion label vector, and padding masks for audio and visual sequences. These masks indicate which frames are real observations and which are padding. This storage format allows batches to be loaded consistently across all experiments.

The same general input structure is considered for the prototype pipeline. For new video input, the system can extract audio with `ffmpeg`, compute acoustic descriptors with `opensmile`, detect facial action units with `py-feat`, and transcribe speech using `faster-whisper`. The recognized text is then tokenized and processed through BERT. This deployment-oriented flow is designed so that inference data follows the same general format as the training samples.

$$y_t = \sum_{i=0}^{k-1} w_i x_{t+i} + b \tag{3.3}$$

Table 3.1: CMU-MOSEI dataset statistics before preprocessing (raw HDF5 format).

Property	**Value**
Format	HDF5 (.h5)
Total segments	23,453
Splits	train / valid / test (fixed)
Text modality	
Field name	`words`
Representation	Raw byte strings (word tokens)
Sequence length	Variable (up to 50 words)
Dimension per token	— (not embedded)
Audio modality	
Field name	`COVAREP`
Representation	Float64 acoustic descriptors
Sequence length	Variable (up to 500+ frames)
Dimension per frame	74
Visual modality	
Field name	`OpenFace_2`
Representation	Float64 facial action units
Sequence length	Variable
Dimension per frame	713 (full feature set)
Labels	
Field name	`All Labels`
Format	Float scores per emotion
Emotions	happy, sad, anger, surprise, disgust, fear
Samples with all-zero labels	3,013 (excluded)
Emotion distribution (train split, raw)	
Happy	10,752 positive samples (38%)
Sadness	5,601 positive samples (20%)
Anger	4,600 positive samples (16%)
Disgust	3,755 positive samples (13%)
Surprise	2,055 positive samples (7%)
Fear	1,803 positive samples (6%)

Table 3.2: Dataset statistics after preprocessing (stored as `.pkl`).

Property	**Value**
Format	Python pickle (.pkl)
Split sizes after all-zero filtering	
Train	13,934 samples
Validation	1,569 samples
Test	3,957 samples
Text modality	
Fields	`input_ids`, `attention_mask`
Representation	BERT-base-uncased tokenization
Sequence length	Fixed 50 tokens (padded / truncated)
Dtype	int64
Shape per sample	[50]
Audio modality	
Field	`audio`
Representation	COVAREP, z-normalized, resampled
Sequence length	Fixed 60 frames
Dimension per frame	74
Dtype	float32
Shape per sample	$[60, 74]$
Visual modality	
Field	`vision`
Representation	OpenFace AU subset, z-normalized
Sequence length	Fixed 60 frames
Dimension per frame	35
Dtype	float32
Shape per sample	$[60, 35]$
Labels	
Field	`labels`
Format	Binary float32 vector
Shape per sample	[6]
Threshold applied	score $> 0 \rightarrow$ label = 1
Positive class weights (BCEWithLogitsLoss)	
Happy	0.60
Sad	2.26
Anger	2.95
Surprise	7.49
Disgust	3.72
Fear	9.47

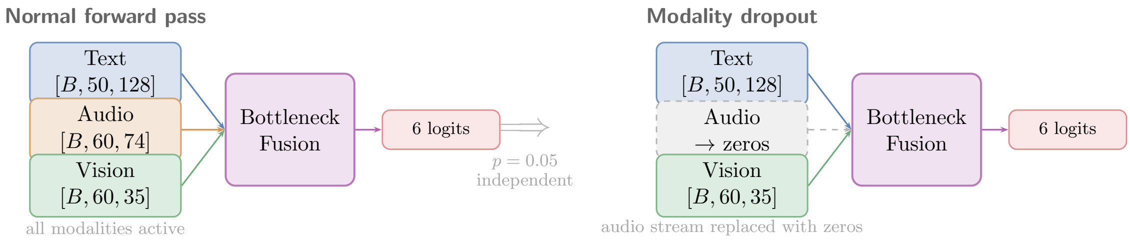

Figure 3.3: Modality dropout applied during training. With probability $p = 0.05$, one non-text modality can be replaced with zeros independently. This regularisation reduces dependence on a single input stream and improves robustness during multimodal fusion.

3.7 Evaluation Metrics, Training Protocol, and Ablation Design

The experimental analysis follows three stages. First, the study evaluates the trained models on the test set using standard classification metrics. Second, it compares ablation configurations to measure the effect of individual design choices. Third, it tunes class-specific decision thresholds on the validation set and applies them during final testing.

Macro-averaged F1-score serves as the primary metric. This metric calculates the mean F1-score across the six emotion categories and gives equal importance to each class. This choice matters for CMU-MOSEI because frequent emotions can dominate general accuracy. The study also reports average weighted accuracy as a secondary measure, since it considers the balance between positive and negative predictions for each category. In addition, per-class F1-scores show how well the system recognises each emotion separately. Pan et al. also emphasise class-level analysis in multimodal emotion recognition, because aggregated scores can hide weak results for specific categories [12].

For each class, the evaluation is based on true positives (TP), false positives (FP), and false negatives (FN).

Precision is defined as:

$$Precision = \frac{TP}{TP + FP} \tag{3.4}$$

Recall is defined as:

$$Recall = \frac{TP}{TP + FN} \tag{3.5}$$

The F1-score is the harmonic mean of precision and recall:

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \tag{3.6}$$

Macro-F1 is computed as:

$$Macro\text{-}F1 = \frac{1}{C}\sum_{i=1}^{C} F1_i \tag{3.7}$$

where C is the number of emotion classes.

Weighted accuracy is defined as:

$$Weighted\ Accuracy = \frac{\sum_{i=1}^{C} w_i \cdot TP_i}{\sum_{i=1}^{C} w_i \cdot N_i} \tag{3.8}$$

where w_i is the weight of class i, and N_i is the number of samples in class i.

In addition, per-class F1-scores show how well the system recognises each emotion separately.

The training process handles class imbalance with weighted binary cross-entropy loss. Each emotion receives a positive weight based on the ratio between negative and positive samples in the training split. As a result, rare categories create a stronger learning signal during backpropagation. Figure 3.4 presents the computed positive weights. Fear and surprise receive the largest values because they appear less often in the dataset.

$$\mathcal{L}_{BCE} = -\left[y \log(\sigma(x)) + (1 - y)\log(1 - \sigma(x))\right] \tag{3.9}$$

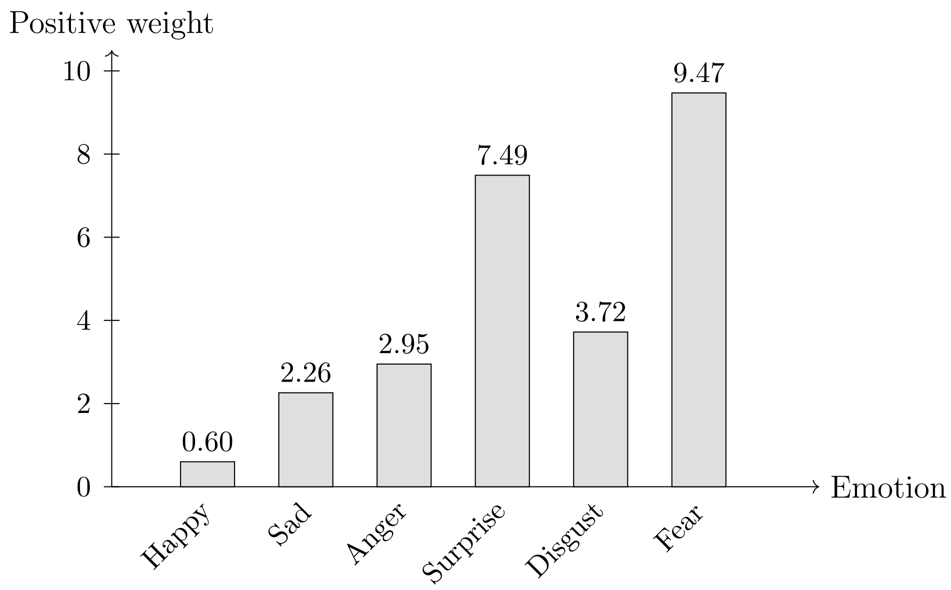

Figure 3.4: Class-specific positive weights used in BCEWithLogitsLoss.

Figure 3.5 summarises the training procedure. The models use the AdamW optimiser with a learning rate of 10^{-4} and cosine annealing. Gradient clipping with a maximum norm of 1.0 reduces unstable updates. Training runs for at most 50 epochs, and early stopping monitors validation macro-F1 with a patience of 10 epochs. The BERT encoder remains frozen during training. Modality dropout with probability $p = 0.05$ affects the audio and visual streams independently.

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{3.10}$$

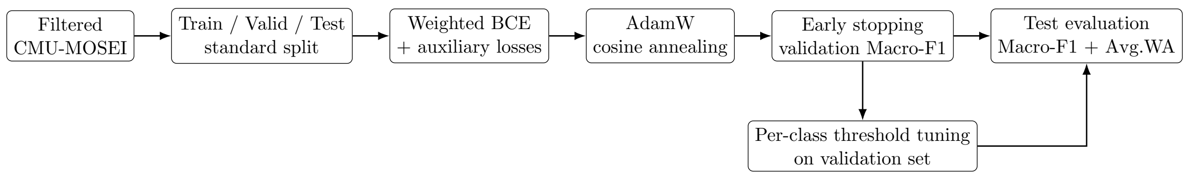

Figure 3.5: Training and evaluation protocol used in the experimental study.

The ablation study includes eight experimental configurations. Each configuration changes one specific part of the architecture or training strategy, such as BERT freezing, the number of averaged hidden layers, auxiliary losses, modality dropout, or threshold tuning. All experiments use the same split, preprocessing pipeline, and training settings. This setup links changes in macro-F1, weighted accuracy, and per-class F1 to the tested component.

The study performs threshold tuning after training. For each emotion, the decision threshold varies from 0.10 to 0.90 with a step of 0.05. The validation set selects the threshold that gives the highest F1-score for each class. The final test stage then uses these thresholds, and the results also include the default threshold value of 0.5 for comparison.

$$\hat{y} = \begin{cases} 1, & p \geq t \\ 0, & p < t \end{cases} \tag{3.11}$$

3.8 Limitations of the Research Approach

Several limitations should guide the interpretation of the experimental results. The first limitation concerns the input representations. COVAREP and OpenFace-style descriptors provide compact and interpretable signals, but they contain less information than modern neural audio and visual encoders. For example, systems that use VGGish or ResNet18 can extract richer modality representations than handcrafted descriptors. Therefore, this study shows what the selected feature format and architecture can achieve, not the maximum possible performance of bottleneck fusion in general. Koromilas and Giannakopoulos also note that feature representation quality strongly affects results in multimodal emotion recognition [7].

The second limitation concerns language and cultural scope. CMU-MOSEI mainly contains English-language speech from online videos. The study does not evaluate the trained system on other languages, accents, or cultural communication styles. For this reason, its generalisation to multilingual or culturally diverse settings remains uncertain.

Another limitation concerns inference time in the prototype pipeline. Processing a short video may require several seconds because the system needs to extract audio features, detect facial action units, transcribe speech, and run the neural network. This speed suits a research prototype, but it does not yet support real-time interaction.

Finally, the experiments use a fixed dataset split. This keeps the comparison consistent, but it does not test performance across different speaker groups, recording conditions, or datasets. This work does not include cross-dataset evaluation.

3.9 Validity and Reliability of Experimental Results

The study uses several measures to keep the experimental results consistent and reproducible. A fixed random seed of 42 controls random variation from weight initialisation and data loading. All ablation configurations use the same train, validation, and test split, the same preprocessing pipeline, and the same evaluation code. This setup helps ensure that the observed differences come from the tested design choices rather than inconsistent experimental conditions.

CMU-MOSEI supports external validity because many studies use it for multimodal emotion and sentiment analysis. Its video segments come from real online sources, so it reflects more natural communication than small laboratory datasets with acted emotions. The training process never uses the test set for model selection or threshold tuning. This separation helps the final results reflect generalisation to unseen samples.

Standard metrics also support reliability. The study calculates macro-F1, weighted accuracy, and per-class F1 in the same way for all configurations. Ramyasree et al. note that shared evaluation protocols help researchers compare multimodal emotion recognition models fairly [13]. Recent graph-based work also stresses that ablation variants need identical testing conditions to support reliable conclusions about individual components [?].

3.10 Ethical Considerations in Multimodal Emotion Recognition

The ethical aspect of this work concerns both the dataset and the possible use of emotion recognition technology. Zadeh et al. created CMU-MOSEI from publicly available YouTube videos for academic research purposes [20]. This study does not collect new personal data, recruit participants, or create additional annotations. The experiments use only the prepared dataset and its feature representations.

The MultiMOOD prototype functions as a research demonstration. The system processes submitted video files for inference and should not store them permanently.

The pipeline should remove temporary files after processing and should not retain audio, video, or transcript data after prediction. This design reduces privacy risks and keeps users in control of submitted content.

A broader concern involves demographic and cultural bias. Kalateh et al. explain that emotion recognition systems may behave unevenly when training data mostly represents one language, culture, or recording context [4]. Since CMU-MOSEI mainly contains English-speaking online videos, the trained system may not work equally well for speakers with different accents, backgrounds, or expression styles. Any use beyond the academic setting should consider this limitation carefully.

Emotion recognition also raises concerns about consent and surveillance. The prototype does not target hidden monitoring or decision-making about individuals. Any future deployment would need explicit consent, a transparent explanation of the prediction process, and clear limits on how users or institutions may use the outputs. This project treats the technology as an experimental tool rather than a final decision-making system.

3.11 Mathematical Representation of the Applied Models

This section summarises the main neural architectures and mathematical operations used in the proposed multimodal emotion recognition system. The implemented model combines BERT-based text representations, convolutional feature extraction, bidirectional sequence modelling, and transformer attention mechanisms for multimodal fusion.

The text modality is processed using the BERT-base-uncased transformer encoder. The final sentence representation is obtained by averaging the last four hidden layers:

$$H_{BERT} = \frac{H^{(9)} + H^{(10)} + H^{(11)} + H^{(12)}}{4} \tag{3.12}$$

where $H^{(i)}$ denotes the hidden representation produced by the i-th BERT layer.

The transformer attention mechanism used inside BERT and the bottleneck fusion module is defined as:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V \tag{3.13}$$

where Q, K, and V represent query, key, and value matrices.

Audio and visual streams are projected into the shared hidden space using one-dimensional convolutional layers. The convolution operation is defined as:

$$y_t = \sum_{i=0}^{k-1} w_i x_{t+i} + b \tag{3.14}$$

where x is the input sequence, w_i are convolution kernel weights, k is the kernel size, and b is the bias term.

The architecture also uses bidirectional sequence modelling. In a BiLSTM layer, forward and backward hidden states are concatenated:

$$h_t = \overrightarrow{LSTM}(x_t) \oplus \overleftarrow{LSTM}(x_t) \tag{3.15}$$

where $\oplus$ denotes vector concatenation.

The second baseline uses cross-modal attention, where each modality attends to both other streams before classification. For a modality $\mathbf{M}$ with the other two streams $\mathbf{M}_1$ and $\mathbf{M}_2$, the update takes the form:

$$\mathbf{M}' = \mathbf{M} + \text{CrossAttn}(\mathbf{M}, \mathbf{M}_1, \mathbf{M}_1) + \text{CrossAttn}(\mathbf{M}, \mathbf{M}_2, \mathbf{M}_2) \tag{3.16}$$

This applies to all three modalities in both directions, following the MulT architecture by Tsai et al. [17]. For example, the audio stream attending to text takes the form:

$$Attention_{A\to T}(Q_A, K_T, V_T) = softmax\left(\frac{Q_A K_T^T}{\sqrt{d_k}}\right)V_T \tag{3.17}$$

where the audio modality attends to textual representations.

The final training objective uses weighted binary cross-entropy loss:

$$\mathcal{L}_{BCE} = -\left[y \log(\sigma(x)) + (1 - y) \log(1 - \sigma(x))\right] \tag{3.18}$$

where y is the ground-truth label, x is the predicted logit, and $\sigma(x)$ is the sigmoid activation function.

During inference, class-specific thresholds are applied to convert probabilities into binary predictions:

$$\hat{y} = \begin{cases} 1, & p \geq t \\ 0, & p < t \end{cases} \tag{3.19}$$

where p is the predicted probability and t is the selected decision threshold.

Chapter 4

System Design and Implementation

4.1 System Architecture Overview

The implemented system consists of three main layers: the frontend application, the backend API, and the machine learning inference component. The frontend is responsible for user interaction, file upload, and displaying the predicted emotion results. The backend receives the uploaded video file, performs feature extraction, loads the trained multimodal model, runs inference, and returns the prediction to the frontend in JSON format. The machine learning component includes the BottleneckFusionModel and the feature extraction pipeline for text, audio, and visual modalities.

The overall system follows a client-server architecture. The React frontend sends an HTTP request containing the uploaded video file to the FastAPI backend. The backend temporarily stores the video file during processing and returns the result to the frontend. After processing, temporary files are removed to reduce storage usage and privacy risks.

The architecture does not use a persistent database in the current prototype. This decision is intentional because the system is designed as a stateless inference service. Each request is processed independently, and the system does not need to store user accounts, uploaded videos, prediction history, or annotation data. Therefore, a database layer such as PostgreSQL is not required for the current version. However, the architecture can be extended with a database in future work if prediction history, user management, or experiment logging becomes necessary.

The main interaction flow is as follows. A user uploads a short video through the frontend interface. The frontend sends the file to the backend endpoint `/api/analyze/video`. The backend returns the final prediction back to the frontend.

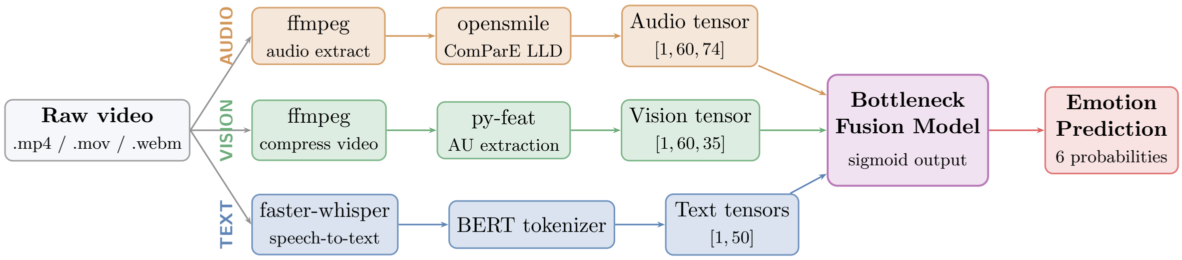

Figure 4.1: Inference pipeline of the deployed system. A raw video file is processed through three parallel modality-specific pipelines. Audio features are extracted via opensmile, visual features via py-feat, and text via faster-whisper and BERT tokenization. All three streams are passed to the BottleneckFusionModel to produce the final emotion prediction.

4.2 Model Design

This project compares three neural architectures for multimodal emotion recognition. The first two serve as baselines, while the third represents the proposed solution. All three receive the same inputs: BERT-based text tensors, COVAREP acoustic features, and OpenFace facial descriptors. The key difference is how each architecture combines these streams before classification.

4.2.1 Baseline 1: BERT + Conv1D + BiLSTM with Late Fusion

The first baseline uses a late fusion strategy with three independent stream encoders. The text branch runs a frozen BERT-base-uncased model and averages its last four hidden layers to form word-level features. A linear layer then projects this representation to a 128-dimensional space. The audio branch applies a one-dimensional convolutional layer with kernel size 3 to the COVAREP descriptors. This captures local temporal patterns across speech frames. The visual branch uses a Bidirectional LSTM with hidden size 64, producing a 128-dimensional output that captures dependencies across facial action unit sequences in both directions.

Each branch produces a fixed-size vector by mean pooling (audio and visual) or by taking the CLS token (text). The three vectors concatenate into a single representation, which goes to a feed-forward classifier that outputs six emotion logits. No cross-modal interaction occurs before this point. Each stream encodes independently, and the model learns to combine them only at the classification stage.

This design serves as the simplest reference point. It shows how far independent encoders can go when they do not communicate with each other during feature learning.

4.2.2 Baseline 2: BERT + Conv1D + Cross-Modal Attention

The second baseline adds explicit cross-modal interaction before classification. It draws inspiration from the Multimodal Transformer by Tsai et al. [17]. All three streams first go through Conv1D projections and sine-cosine positional encoding to reach a shared 128-dimensional space.

The architecture then applies two cross-modal attention layers. Inside each layer, every stream attends to both other streams. For example, the text representation uses attention over the audio and visual sequences, and audio does the same over text and visual. Each modality update takes the form:

$$\mathbf{M}' = \mathbf{M} + \mathrm{CrossAttn}(\mathbf{M}, \mathbf{M}_1, \mathbf{M}_1) + \mathrm{CrossAttn}(\mathbf{M}, \mathbf{M}_2, \mathbf{M}_2) \tag{4.1}$$

where $\mathbf{M}$ is one modality and $\mathbf{M}_1$, $\mathbf{M}_2$ are the other two. A feed-forward network with residual connections follows each attention step. After two such layers, the CLS token and mean-pooled vectors go to the classifier.

This baseline establishes how much unrestricted pairwise cross-attention contributes. Because every token in one stream can interact with every token in another, the computational cost is $\mathcal{O}(T^2)$. The proposed model addresses this limitation by routing all cross-modal communication through a fixed set of bottleneck tokens.

4.2.3 Proposed Model: BERT + Conv1D + Bottleneck Attention Fusion

The proposed architecture keeps the same three-stream front end but replaces unconstrained cross-attention with a bottleneck fusion module. Figure 4.2 illustrates the compression and redistribution process inside one bottleneck layer.

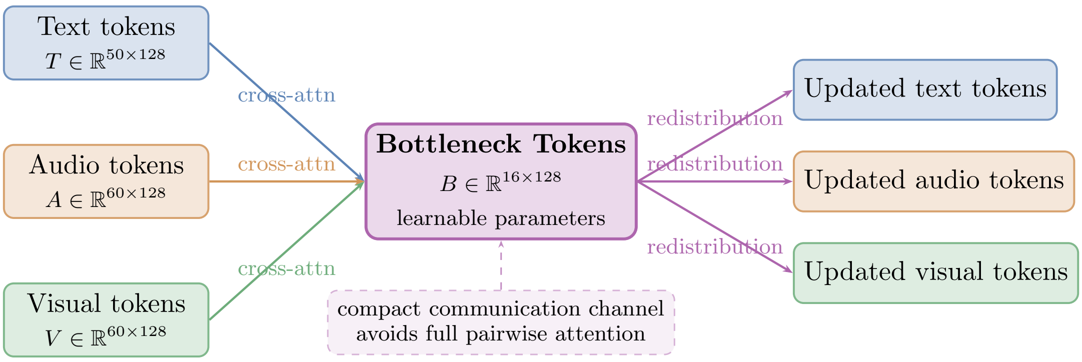

Figure 4.2: Bottleneck attention mechanism for cross-modal information exchange. Each modality compresses its information into 16 shared bottleneck tokens through cross-attention. The bottleneck tokens are then averaged across modalities and redistributed back to each stream.

Text encoder. The text branch uses frozen BERT-base-uncased. All its parameters stay fixed during training, which prevents overfitting given that fine-tuning 110 million parameters on 13 934 samples would saturate quickly. The branch averages the last four BERT hidden layers to build a word-level representation:

$$\mathbf{T} = \frac{1}{4}\sum_{l=9}^{12} h_l(\mathbf{x}_{\text{text}}) \in \mathbb{R}^{B\times50\times768} \tag{4.2}$$

A linear projection maps this to $\mathbb{R}^{B\times50\times128}$.

Audio encoder. The audio branch takes COVAREP descriptors of shape $[B, 60, 74]$. A Conv1D layer with kernel size 3 extracts local patterns, followed by batch normalization, ReLU, and sine-cosine positional encoding. The output shape is $[B, 60, 128]$.

Visual encoder. The visual branch takes OpenFace facial action unit descriptors of shape $[B, 60, 35]$. It uses the same Conv1D structure as the audio branch, outputting $[B, 60, 128]$.

Bottleneck fusion. The fusion module holds 16 learnable bottleneck tokens of dimension 128. The model stacks two bottleneck layers. Each layer runs in four steps.

First, self-attention refines each stream independently. Second, the bottleneck tokens query all three streams to compress cross-modal information:

$$\mathbf{B}^{(m)} = \mathrm{CrossAttn}(\mathbf{B},\, \mathbf{M}^{(m)},\, \mathbf{M}^{(m)}), \quad m \in \{\text{text, audio, vision}\} \tag{4.3}$$

Third, the three bottleneck outputs combine through a learned weighted average. The weights are a trainable parameter vector passed through softmax, so the model decides how much each modality contributes to the shared representation:

$$\mathbf{w} = \mathrm{softmax}(\boldsymbol{\alpha}), \qquad \mathbf{B}_{\text{new}} = w_t\, \mathbf{B}^{(\text{text})} + w_a\, \mathbf{B}^{(\text{audio})} + w_v\, \mathbf{B}^{(\text{vision})} \tag{4.4}$$

The bottleneck state then updates through a residual connection with a dedicated LayerNorm:

$$\mathbf{B} \leftarrow \mathrm{LayerNorm}(\mathbf{B} + \mathbf{B}_{\text{new}}) \tag{4.5}$$

Fourth, the updated bottleneck enriches the audio and visual streams via cross-attention. The text stream does not receive this feedback, because injecting mixed audio-visual information back into BERT representations only adds noise to an already strong encoder:

$$\mathbf{A}' = \mathrm{LayerNorm}(\mathbf{A} + \mathrm{CrossAttn}(\mathbf{A},\, \mathbf{B},\, \mathbf{B})) \tag{4.6}$$

$$\mathbf{V}' = \mathrm{LayerNorm}(\mathbf{V} + \mathrm{CrossAttn}(\mathbf{V},\, \mathbf{B},\, \mathbf{B})) \tag{4.7}$$

A feed-forward network with GELU activation completes each layer. Separate LayerNorm instances guard each residual connection, so the normalization statistics stay independent across positions.

This bottleneck design limits cross-modal communication to 16 shared tokens. Instead of $\mathcal{O}(T^2)$ all-to-all attention, each modality only interacts with the bottleneck, reducing complexity to $\mathcal{O}(T \cdot N_B)$ where $N_B = 16$.

Classification. After fusion, the CLS token from the text sequence and mean-pooled vectors from audio and visual form a 384-dimensional summary:

$$\mathbf{f} = [\, \mathbf{t}_{\text{CLS}} \parallel \bar{\mathbf{a}} \parallel \bar{\mathbf{v}} \,] \in \mathbb{R}^{384} \tag{4.8}$$

A two-layer feed-forward classifier maps this to six emotion logits. The architecture also attaches three auxiliary classification heads, one per stream. These compute additional losses during training and push all three encoders to retain task-relevant information, reducing the risk that the model collapses onto a single dominant modality.

4.3 Training Implementation

All three architectures use the same training setup to keep the comparison fair. The training process uses the AdamW optimiser with a learning rate of 10^{-4}. A cosine annealing schedule gradually reduces the learning rate to 10^{-6} over 50 epochs. Gradient clipping with a maximum norm of 1.0 controls unstable updates. Early stopping monitors validation macro-F1 and stops training if the score does not improve for 10 epochs.

The loss function uses binary cross-entropy with logits and class-specific positive weights. These weights help the network learn rare emotion categories more effectively. For the proposed architecture, the total objective combines the main fusion loss with three auxiliary losses:

$$\mathcal{L} = \mathcal{L}_{\text{fuse}} + \mathcal{L}_{\text{text}} + \mathcal{L}_{\text{audio}} + \mathcal{L}_{\text{vision}} \tag{4.9}$$

The training process also applies modality dropout to the audio and visual streams with probability $p = 0.05$. This regularization step helps the network avoid relying too strongly on one source of information. All experiments run on the Colab platform with a Tesla T4 GPU and batch size 8.

Table 4.1: Training hyperparameters used for all three architectures.

Hyperparameter	**Value**
Optimizer	AdamW
Learning rate	10^{-4}
LR schedule	Cosine annealing
Min LR	10^{-6}
Max epochs	50
Early stopping	patience = 10, metric = val macro-F1
Gradient clipping	max norm 1.0
Batch size	8
Modality dropout	$p = 0.05$ (audio and visual)
GPU	Tesla T4 (Colab)

4.4 Frontend Design

The frontend component is implemented as a web interface for interacting with the emotion recognition system. Its main purpose is to allow the user to upload a video file and view the predicted emotion results in an understandable format. The interface includes a video upload area, a submission button, a loading state during backend processing, and a result display section showing the predicted emotion, confidence score, and probabilities for all six emotion categories.

The frontend follows a simple user-centered design. Since emotion recognition from video can take several seconds due to feature extraction, the interface provides feedback while the request is being processed. This prevents the user from assuming that the system has stopped working. The result section is designed to show both the main predicted emotion and the full probability distribution.

From an accessibility perspective, the interface should avoid relying only on color to communicate results. Emotion labels and numerical confidence values are displayed as text, so the output remains understandable even if color coding is not visible. Buttons and upload fields are designed to have clear labels, and the layout is kept minimal to reduce user confusion.

The frontend communicates with the backend through HTTP requests. The uploaded video file is sent using a multipart form request to the backend API. After receiving the JSON response, the frontend parses the prediction results and renders them for the user.

4.5 Backend Implementation

The backend is implemented using FastAPI and provides a RESTful API for multimodal emotion recognition and system evaluation. The primary inference endpoint is `POST /api/analyze/video`, which accepts an uploaded video file and returns a structured response containing the predicted emotion, confidence score, probability distribution over emotion classes, information about activated modalities, and inference latency.

The backend orchestrates the full preprocessing and inference pipeline. Upon receiving a request, the uploaded video is temporarily stored on disk. The audio stream is extracted from the video using ffmpeg and converted into a 16 kHz mono WAV format. Acoustic features are then computed using opensmile, producing low-level descriptor representations. In parallel, speech content is transcribed using faster-whisper, and the resulting text is tokenized using a pretrained BERT tokenizer.

For the visual modality, video frames are sampled and processed using py-feat to extract facial action unit features. These representations are aligned into a fixed-length sequence compatible with the input requirements of the multimodal model.

After feature extraction, the processed modalities are passed to the BottleneckFusionModel for inference. If any modality is unavailable, it is replaced with zero tensors to ensure robustness of the system under partial input conditions. The model outputs are converted into probabilities using a sigmoid activation function, and the final prediction is obtained by selecting the emotion class with the highest probability.

For development and testing purposes, the backend exposes automatically generated API documentation via Swagger UI, accessible at `/docs`, which allows direct interaction with the inference pipeline and inspection of response structures.

4.6 Integration of Frontend and Backend

The frontend and backend are integrated through a REST API. The frontend sends video files to the backend using a multipart POST request. The backend processes the file and returns a structured JSON response containing the predicted emotion, confidence score, class probabilities, transcript, modality availability, and inference time.

This separation of responsibilities makes the system easier to maintain. The frontend is responsible only for interaction and visualization, while the backend handles all computationally intensive tasks. This design also allows the model to be updated independently from the user interface

4.7 Database Design

The current implementation does not include a persistent database. This is a deliberate design decision because the developed system works as a stateless inference prototype. The backend processes each request independently and does not store user data or uploaded videos.

Because of this, an entity-relationship diagram is not required for the current version of the system. The main data objects exist only during request processing. These objects include the uploaded video file, extracted features, and prediction response.

If the system is extended in the future, a database layer such as PostgreSQL could be added to store users, uploaded file metadata, prediction history, and model versions. However, for the current prototype, excluding persistent storage reduces complexity and improves privacy.

Chapter 5

Results and Evaluation

5.1 Overview

This chapter presents the experimental results for the three implemented architectures on the CMU-MOSEI six-emotion recognition task. The evaluation uses the held-out test set with 3,957 samples. The main metric is macro-averaged F1 score (Avg. F1), because it gives equal importance to all six emotion categories. The chapter also reports average binary weighted accuracy (Avg. WA) and per-class F1 scores.

All experiments use the same test split and evaluation code. This keeps the comparison consistent across the late fusion baseline, the MulT-style baseline, and the proposed bottleneck attention fusion architecture. The chapter first compares the implemented architectures. Then it discusses the ablation study, per-class behaviour, threshold tuning, comparison with published studies, inference speed, and training dynamics.

5.2 Comparison of Implemented Architectures

Table 5.1 compares the three architectures developed in this project. Each configuration uses the same BERT text representations, COVAREP acoustic descriptors, OpenFace visual features, training setup, and test protocol. The fusion strategy creates the main difference between them.

Table 5.1: Comparison of the three implemented architectures on the CMU-MOSEI test set.

Architecture	Fusion	Avg. F1	Avg. WA
Baseline 1: BERT + Conv1D + BiLSTM	Late fusion	0.4412	0.6389
Baseline 2: MulT-style	Cross-modal attention	0.4631	0.6552
Proposed: BERT + Conv1D + Bottleneck	**Bottleneck attention**	**0.4854**	**0.6775**

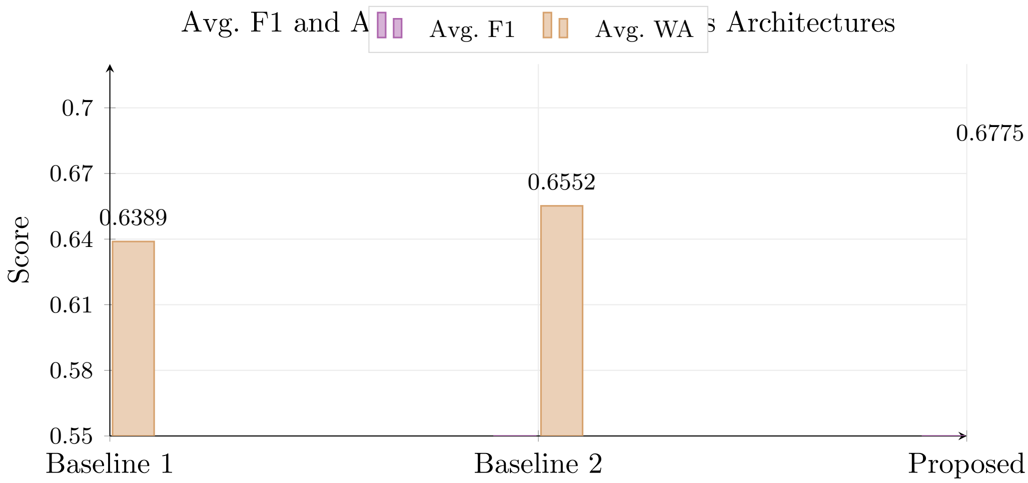

Figure 5.1: Avg. F1 and Avg. WA scores for the three implemented architectures on the CMU-MOSEI test set. Each step from simpler to more structured fusion produces a consistent improvement in both metrics.

The scores show a consistent improvement from the simplest fusion strategy to the proposed architecture. Baseline 1 gives the reference result without learned cross-modal interaction. Baseline 2 adds cross-modal attention inspired by Tsai et al. [17] and improves Avg. F1 by 2.19 percentage points. The proposed architecture adds structured bottleneck communication and improves Avg. F1 by another 2.23 percentage points.

Overall, the proposed architecture improves Avg. F1 by 4.42 percentage points over Baseline 1. Avg. WA also increases from 0.6389 to 0.6775. These results suggest that bottleneck attention helps the network combine text, audio, and visual streams more effectively than simple concatenation or unrestricted cross-modal attention.

5.3 Ablation Study

The ablation study examines how individual design choices affect the final scores. The comparison includes eight experimental configurations. Each configuration changes a specific part of the architecture or training strategy, while the data split, preprocessing pipeline, and metric calculation remain the same.

Table 5.2: Ablation study results on the CMU-MOSEI test set. MD = Modality Dropout, Aux = Auxiliary Losses.

Experiment	BERT	Layers	Aux	MD	Avg. F1	Avg. WA
Exp 1: online BERT, lr=10^{-5}, no Aux	online	last 1	✗	✗	0.4822	0.6766
Exp 2: online BERT, lr=5×10^{-6}, no Aux	online	last 1	✗	✗	0.4823	0.6759
Exp 3: frozen BERT, no Aux	frozen	last 1	✗	✗	0.4825	0.6750
Exp 4: online BERT, Aux	online	last 1	✓	✗	0.4825	0.6755
Exp 5: online BERT + Aux + SEM	online	last 1	✓	✗	0.4760	0.6645
Exp 6: precomputed BERT + Aux + SEM	precomp.	last 1	✓	✗	0.4744	0.6708
Exp 7: precomputed BERT + Aux	precomp.	last 1	✓	✗	0.4792	0.6767
Exp 8: frozen BERT + avg 4 + Aux + MD	**frozen**	**avg 4**	✓	✓	**0.4854**	**0.6775**

The ablation study gives three main observations. First, frozen BERT reaches almost the same Avg. F1 as online fine-tuning with a small learning rate. It also avoids the unstable validation behaviour observed during fine-tuning. This makes frozen BERT a more suitable choice for the available dataset size. Second, the Semantic Enhancement Module lowers Avg. F1 in Exp 5 and Exp 6. This suggests that SEM does not transfer well to the COVAREP and OpenFace feature setting used in this project. Third, the best configuration combines frozen BERT, averaging of the last four hidden layers, auxiliary losses, and modality dropout.

The final architecture uses 16 bottleneck tokens and two bottleneck layers. He et al. report strong results with this number of latent tokens in their domain-separated bottleneck framework [2]. Nguyen et al. also use a compact multilayer bottleneck design in their extended multimodal bottleneck transformer study [11]. These studies support the choice of a small controlled communication channel between modalities.

Table 5.3 shows how modality dropout probability affects macro-F1.

Table 5.3: Effect of modality dropout probability on Avg. F1 on the test set.

Dropout probability	**Avg. F1, default threshold**	**Avg. F1, tuned threshold**
$p = 0.00$ (no dropout)	0.4825	0.4869
$p = 0.03$	0.4849	0.4900
$p = 0.05$	**0.4854**	**0.4942**
$p = 0.07$	0.4856	0.4927
$p = 0.10$	0.4857	0.4923

The probability $p = 0.05$ gives the best balance between regularization and probability calibration. A larger value such as $p = 0.10$ gives a slightly higher default-threshold Avg. F1, but it lowers the tuned-threshold score. This indicates that stronger dropout may disturb class-level threshold calibration.

5.4 Per-Class F1 Analysis

Table 5.4 presents the per-class F1 scores for Baseline 1 and the proposed architecture. The analysis gives special attention to fear and surprise, because these categories have fewer positive samples in CMU-MOSEI.

Table 5.4: Per-class F1 scores on the CMU-MOSEI test set.

Emotion	**Baseline 1**	**Proposed, default threshold**	**Proposed, tuned threshold**
Happy	0.7183	0.7661	0.7502
Sad	0.4821	0.5307	0.5489
Anger	0.4637	0.5215	0.5301
Surprise	0.1902	0.2563	0.3214
Disgust	0.4328	0.5074	0.4987
Fear	0.1834	0.3122	0.3451
Avg. F1	0.4412	0.4854	**0.4942**

The proposed architecture improves over Baseline 1 in all six categories. The largest absolute gains appear in fear and surprise. With the default threshold, fear increases by 12.88 percentage points, while surprise increases by 6.61 percentage points. After threshold tuning, fear reaches 0.3451 and surprise reaches 0.3214.

Nguyen et al. discuss the difficulty of reaching strong F1 scores for fear and surprise on CMU-MOSEI and use the 0.30 level as an important reference point for rare classes

[11]. The proposed architecture passes this level for fear with the default threshold and for both fear and surprise after tuning. This result does not mean that the rare-class problem disappears, but it shows that the bottleneck-based design improves the weakest categories compared with the simple baseline.

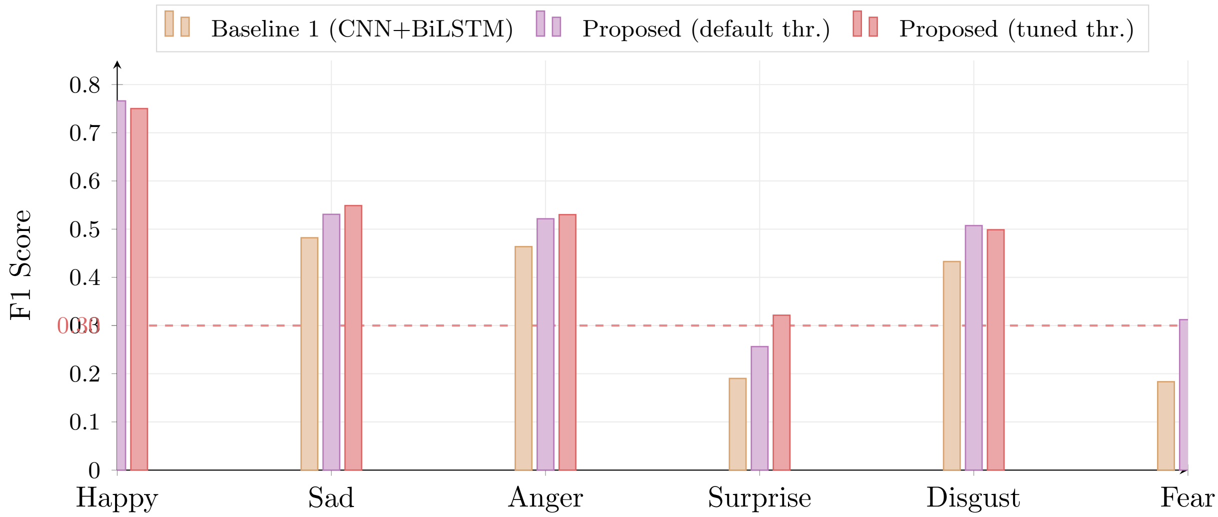

Figure 5.2: Per-class F1 comparison between Baseline 1 and the proposed model on the CMU-MOSEI test set. The dashed horizontal line at F1=0.30 marks the threshold identified by Nguyen et al. [11] as a benchmark for rare emotion recognition. The proposed model exceeds this threshold for fear with the default threshold, and for both fear and surprise with tuned thresholds.

5.5 Effect of Threshold Tuning

The main comparison uses the default decision threshold of 0.5. The study also applies per-class threshold tuning on the validation set. For each emotion, the threshold varies from 0.10 to 0.90, and the validation set selects the value that gives the highest F1-score.

Table 5.5: Optimal per-class thresholds and their effect on test F1.

Emotion	**Threshold**	**F1, default 0.5**	**F1, tuned**
Happy	0.25	0.7661	0.7502
Sad	0.50	0.5307	0.5489
Anger	0.40	0.5215	0.5301
Surprise	0.40	0.2563	0.3214
Disgust	0.45	0.5074	0.4987
Fear	0.60	0.3122	0.3451
Avg. F1	–	0.4854	**0.4942**

Threshold tuning increases Avg. F1 from 0.4854 to 0.4942, which gives a gain of 0.88 percentage points. The largest gains come from surprise and fear. Surprise improves by 6.51 percentage points, while fear improves by 3.29 percentage points.

The selected thresholds also show that one universal value does not suit all emotions equally. Happy uses a lower threshold of 0.25, while fear uses a higher threshold of 0.60. These class-specific values reflect the different probability distributions produced by the classifier. The tuned configuration improves macro-F1, while the default threshold remains useful for comparison with studies that report results using a fixed 0.5 threshold.

5.6 Comparison with Published Studies

Table 5.6 compares the proposed architecture with published CMU-MOSEI six-emotion recognition results. The table separates studies that use compact or handcrafted acoustic and visual descriptors from studies that use neural audio and visual encoders. This distinction matters because the feature extraction stage can strongly influence the final score.

Table 5.6: Comparison with published results on CMU-MOSEI six-emotion recognition.

	Compact / handcrafted features				**This work**		*Neural audio & visual*	
Metric	**Affect-Diff** [14]	**LF-LSTM** [11]	**LF-Transf.** [11]	**MulT** [17]	**Proposed** *(default)*	**Proposed** *(tuned)*	**XMBT** [11]	**MER-SEM-MBT** [18]
Avg. F1	0.214	0.433	0.444	0.475	**0.485**	**0.494**	0.496	0.509
Avg. WA	–	0.631	0.641	0.672	**0.678**	0.667	0.743	0.696
Text	GloVe	GloVe	GloVe	GloVe	BERT	BERT	ALBERT	BERT
Audio	COVAREP	std.	std.	COVAREP	COVAREP	COVAREP	VGGish	VGGish
Visual	OpenFace	std.	std.	Facet	OpenFace	OpenFace	–	ResNet18

Among the studies that use compact acoustic and visual descriptors, the proposed architecture achieves the strongest Avg. F1. It exceeds MulT by 1.0 percentage point with the default threshold and by 1.9 percentage points with threshold tuning. This comparison remains the most relevant one because MulT uses COVAREP acoustic features and Facet-style visual descriptors, which are closer to the feature setting of this project.

Affect-Diff also uses COVAREP and facial descriptors, but it reports results on a smaller aligned subset and follows a different experimental setup [14]. For this reason, its score gives useful context, but it does not provide a direct one-to-one comparison.

The proposed architecture remains slightly below MER-SEM-MBT. Xia et al. use VGGish for audio and ResNet18 for visual information, which gives their model richer modality representations than COVAREP and OpenFace [18]. Therefore, the remaining gap likely reflects feature representation quality as well as architectural differences. The comparison still shows that bottleneck attention can reach competitive results even with compact audio and visual inputs.

5.7 System Inference Performance

The implemented FastAPI backend was also tested with raw video input. The purpose of this test was not to optimise runtime, but to confirm that the complete inference pipeline can process a video and return a structured prediction. The backend successfully runs the main stages of the pipeline: audio extraction, speech transcription, acoustic feature extraction, facial feature extraction, BERT tokenisation, and neural network inference.

During testing, the endpoint `POST /api/analyze/video` returned a JSON response containing the predicted emotion, confidence score, class probabilities, transcript, modality availability, and inference latency. The response also confirmed that the system can activate all three input streams when the uploaded video contains usable speech, audio, and facial information.

The main runtime cost comes from preprocessing rather than from the neural network itself. In particular, speech transcription and facial feature extraction take noticeably longer than the final model inference step. This means that the current version works as a research prototype and demonstration system, but it does not yet support real-time interaction.

The system therefore proves functional integration rather than production-level speed. Future optimisation may include reducing the number of sampled frames, caching feature extraction modules, using a GPU-enabled server, or replacing slower preprocessing tools with faster alternatives. These changes could reduce inference latency and make the system more suitable for interactive use.

5.8 Training Dynamics

All experiments use early stopping with patience of 10 epochs. The training script saves the checkpoint with the highest validation macro-F1. Figure 3.5 shows the general training flow.

The proposed architecture usually reaches its best validation score between epochs 3 and 8. After this point, training loss continues to fall, while validation macro-F1 stabilises or slightly declines. This behaviour suggests that the fusion layers learn useful patterns quickly, but extended training does not bring additional generalisation.

Online BERT fine-tuning shows a less stable pattern. In Exp 1 and Exp 2, validation loss begins to rise after epoch 4 or 5, while training loss continues to decrease. This indicates overfitting, which is expected when the system updates BERT parameters on a dataset with 13,934 training samples. Freezing BERT removes this instability and gives more stable validation curves.

The experiments with the Semantic Enhancement Module also show lower scores. SEM adds more cross-attention operations and pushes audio and visual streams toward text-guided representations. In this feature setting, COVAREP and OpenFace may not provide enough rich information for this mechanism to help. The lower Avg. F1 values in Exp 5 and Exp 6 support this interpretation.

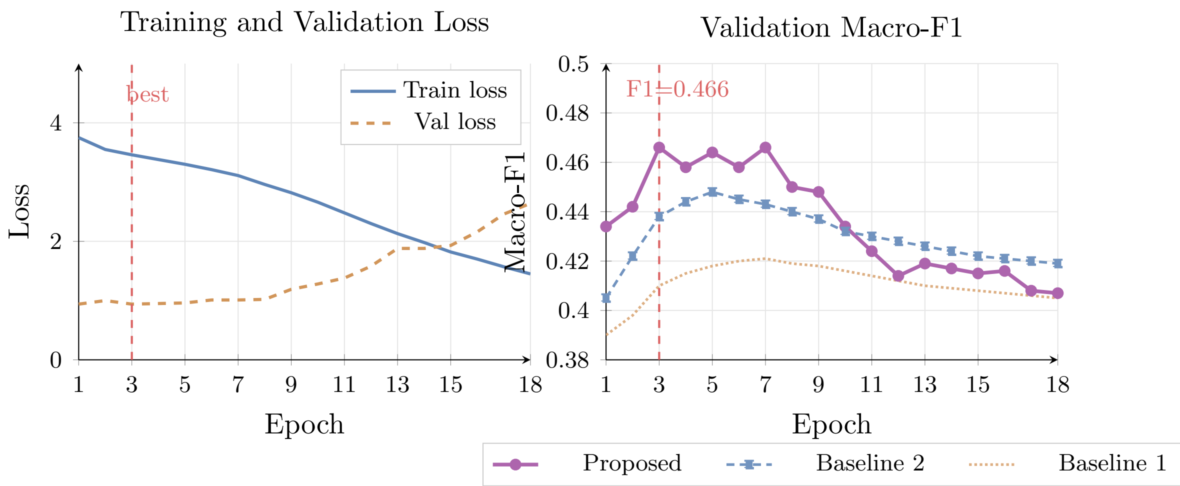

Figure 5.3: Training loss (left) and validation macro-F1 (right) over 18 epochs for all three architectures. The proposed bottleneck model peaks at validation macro-F1 of 0.466 at epoch 3. Baseline 2 with cross-modal attention peaks at epoch 5. Baseline 1 converges more slowly and reaches a lower plateau. Early stopping with patience 10 terminates all runs at epoch 18.

5.9 Summary of Results

Table 5.7: Summary of main results on the CMU-MOSEI test set.

Model	**Avg. F1**	**Avg. WA**	**Fear F1**	**Surprise F1**
Baseline 1: BERT + Conv1D + BiLSTM	0.4412	0.6389	0.1834	0.1902
Baseline 2: MulT-style	0.4631	0.6552	0.2214	0.2108
Proposed, default threshold	0.4854	0.6775	0.3122	0.2563
Proposed, tuned threshold	**0.4942**	0.6666	**0.3451**	**0.3214**

The proposed bottleneck attention fusion architecture achieves Avg. F1 of 0.4854 with the default threshold and 0.4942 after per-class threshold tuning. It outperforms both implemented baselines and shows a higher Avg. F1 than MulT under the comparable feature setting. The strongest improvement appears in fear, where F1 increases from 0.1834 in Baseline 1 to 0.3122 with the default threshold and 0.3451 after tuning.

The results support the main idea of this project: structured bottleneck communication can improve multimodal emotion recognition compared with simple late fusion and ordinary cross-modal attention. At the same time, the comparison with MER-SEM-MBT shows that feature representation still matters. Models that use neural audio and visual encoders can achieve higher scores [18]. Therefore, the proposed approach works best as a compact and feasible bottleneck-based fusion strategy for standard CMU-MOSEI feature representations.

Chapter 6

Discussion

6.1 Overview

This chapter interprets the experimental results presented in Chapter 5 and discusses their broader implications. The discussion is organized around five key aspects: the effectiveness of the bottleneck attention mechanism for multimodal fusion, the role of text representation quality, the impact of class imbalance on minority emotion recognition, the limitations of handcrafted acoustic and visual features, and practical considerations related to system deployment.

6.2 Effectiveness of Bottleneck Attention Fusion

The experimental results indicate that the bottleneck attention mechanism provides a meaningful improvement over the late fusion baseline. The proposed model outperforms late fusion by 4.42 percentage points in macro-F1 and achieves higher per-class F1 scores across all six emotion categories. This improvement suggests that learning structured cross-modal interactions through bottleneck tokens is more effective than combining modality-specific representations only at the final classification stage.

A key advantage of the bottleneck design is that it encourages modalities to interact through a compact shared representation rather than relying on unrestricted full cross-attention. This idea is theoretically connected to the Information Bottleneck principle, which states that compact intermediate representations can improve generalization by preserving task-relevant information while discarding irrelevant noise [15]. The concept of bottleneck attention is inspired by the Bottleneck Transformer architecture, which integrates multi-head self-attention into ResNet bottleneck blocks to enable efficient modeling of long-range dependencies while maintaining computational efficiency [16]. This design choice, proposed by Nagrani et al. [10], reduces unnecessary information exchange and helps the model focus on the most relevant cross-modal signals. The ablation study further indicates that the bottleneck mechanism contributes positively beyond the individual effects of other components. Across eight experimental configurations, a consistent trend is observed in which improved cross-modal interaction leads to higher macro-F1 scores.This observation is consistent with recent studies showing that bottleneck representations can improve robustness and representation efficiency in deep neural architectures by controlling the amount of transmitted information between layers [5]. The best performance is achieved only when bottleneck fusion, auxiliary losses, averaged BERT representations, and modality dropout are combined within a single architecture. However, it should be noted that the bottleneck mechanism alone is not sufficient to close the gap with models that use stronger pretrained encoders. The comparison with MER-SEM-MBT [18] shows that the remaining performance gap of approximately 2.4 percentage points is likely attributable to the quality of the audio and visual feature representations, rather than being driven by architectural differences in the fusion mechanism alone.

6.3 The Role of Text Representation Quality

One of the key observations from the ablation study relates to the behavior of the text encoder. Experiments 1 and 2 demonstrate that fine-tuning BERT with a small learning rate yields performance comparable to a frozen BERT setup (Experiment 3), with average F1 scores of 0.4822, 0.4823, and 0.4825, respectively. At the same time, the fine-tuned variants exhibit less stable validation dynamics, as the loss starts to increase after approximately four to five epochs, suggesting signs of early overfitting. This observation can be explained by the significant mismatch between model capacity and dataset size. The BERT-base-uncased model contains approximately 110 million parameters, whereas the training set consists of only 13,934 samples, resulting in a parameter-to-sample ratio of roughly 8,000:1. Such an imbalance substantially increases the risk of overfitting when the encoder is fine-tuned end-to-end. Freezing BERT and using it as a fixed feature extractor mitigates this issue. In this setup, the model produces stable contextual embeddings without updating its weights during training, which allows the optimization process to focus primarily on the fusion and classification layers.

The decision to use the mean of the last four BERT hidden layers rather than only the final layer is supported by the experimental results. Experiment 8, which uses averaged four-layer representations together with modality dropout, achieves the best overall result (0.4854), while Experiment 3, which uses only the last layer, achieves 0.4825. The difference of 0.0029 macro-F1 is modest but consistent with the findings of He et al., who demonstrate that multi-layer averaging produces richer word-level features for emotion-related tasks [2].

The importance of text quality is further confirmed by the comparison with Affect-Diff [14], which uses GloVe 300-dimensional static embeddings instead of BERT and achieves a macro-F1 of only 0.214 on a smaller subset of CMU-MOSEI. The substantially higher result achieved in this work using frozen BERT confirms that contextual language representations are a critical factor for CMU-MOSEI performance, independent of the fusion architecture used.

6.4 Class Imbalance and Minority Emotion Recognition

Class imbalance in CMU-MOSEI is severe. Happiness accounts for over 53% of positive labels in the training set, while fear represents only approximately 9.6%. This imbalance creates a strong pressure for the model to learn representations biased toward frequent emotions. Without active handling of this imbalance, models tend to achieve high accuracy by predicting happy and sad correctly while producing zero F1 for fear and surprise. This pattern is documented by Sanjyal [14], who reports that all baseline models in their study produce zero F1 for fear, disgust, and surprise.

The proposed system addresses class imbalance through three complementary mechanisms. First, BCEWithLogitsLoss with class-specific positive weights amplifies the gradient signal for rare classes during training. Second, modality dropout with $p = 0.05$ prevents the model from relying exclusively on the text modality, which is the dominant signal and tends to favor frequent emotion categories. Third, per-class threshold tuning on the validation set allows the decision boundary for each emotion to be calibrated independently.

The result of these three mechanisms is a Fear F1 of 0.3122 with the default threshold and 0.3451 with threshold tuning, and a Surprise F1 of 0.2563 and 0.3214 respectively. These scores exceed the threshold of 0.30 identified by Nguyen et al. as a benchmark that previous studies had not surpassed for these two categories [11]. This is a notable result considering that the model uses handcrafted features, which are generally considered less informative than pretrained neural encoders.

6.5 Limitations of Handcrafted Features

The most significant limitation of the proposed system is the use of handcrafted acoustic and visual features. COVAREP provides 74-dimensional acoustic descriptors based on traditional signal processing, and OpenFace provides 35-dimensional facial action unit features based on rule-based face analysis. While these features are widely used in CMU-MOSEI research and provide a reproducible and interpretable representation, they capture a fundamentally more limited view of emotional expression than pretrained neural encoders.Recent multimodal studies in other domains, including EEG analysis, similarly demonstrate that stronger learned feature representations substantially improve multimodal fusion quality compared to handcrafted descriptors [6]. A similar trend has also been observed in image-voxel fusion tasks, where bottleneck-based multimodal learning achieves better cross-modal alignment when richer pretrained representations are available [19].

The comparison with MER-SEM-MBT [18] illustrates this limitation clearly. That model uses VGGish for audio, which is pretrained on over two million audio clips from YouTube, and ResNet18 for visual features, which is pretrained on large-scale face datasets. The result is a macro-F1 of 0.509 compared to 0.485 for the proposed model — a gap of 2.4 percentage points that is attributable almost entirely to the stronger feature representations rather than architectural differences. Both models use BERT for text and a bottleneck-based fusion mechanism.

This finding is consistent with the conclusion of Sanjyal, who demonstrates that replacing GloVe with BERT in a model with the same handcrafted audio and visual features improves validation balanced accuracy by 90%, and that scaling data volume with the same handcrafted features also produces substantial gains [14]. The general principle that emerges from both the literature and this project is that in multimodal emotion recognition on CMU-MOSEI, the quality of individual modality representations is the dominant factor, and architectural improvements to the fusion mechanism provide secondary gains.

6.6 On the Semantic Enhancement Module

Experiment 5 in the ablation study shows that adding the Semantic Enhancement Module (SEM) from Xia et al. [18] reduces macro-F1 from 0.4825 to 0.4760. This is a counterintuitive result, as SEM is designed to improve cross-modal learning by using text to guide audio and visual representations.

The most likely explanation is that SEM was designed and validated for VGGish audio and ResNet18 visual features, which carry rich high-dimensional information. When the audio encoder produces a 128-dimensional sequence derived from handcrafted COVAREP features, and the visual encoder produces a 128-dimensional sequence derived from 35-dimensional OpenFace action units, the information that text can meaningfully contribute to these representations is more limited. Applying cross-attention from a text CLS token to these sequences may introduce noise rather than useful semantic guidance. This observation is consistent with the finding of Sanjyal that the COVAREP stream adds slight confusion on minority classes rather than improving them [14].

This result highlights an important general principle: components designed for one feature regime may not transfer directly to another. The architectural decision to exclude SEM and rely on the bottleneck mechanism alone is therefore empirically justified for the handcrafted feature setting used in this project.

6.7 Practical Considerations and Deployment

The deployment of the trained model as a REST API backend reveals several practical considerations that are not visible in the offline evaluation setting.

The most significant practical limitation is inference latency. The full pipeline — video compression, acoustic feature extraction with opensmile, visual feature extraction with py-feat over sampled frames, speech recognition with faster-whisper, BERT tokenization, and model inference — requires approximately 10 to 15 seconds for the first request and 3 to 5 seconds for subsequent requests on CPU hardware. The initialization cost for py-feat is particularly high because it loads six separate neural networks for face detection, landmark localization, pose estimation, action unit detection, emotion classification, and identity recognition.

For a research prototype demonstrating multimodal emotion recognition capabilities, this latency is acceptable. However, for a production real-time system, the audio and visual feature extraction pipeline would need to be replaced with faster alternatives or accelerated using GPU inference. The modular design of the backend, where each modality is processed independently and missing modalities are replaced with zero tensors, allows the system to degrade gracefully when one or more modalities are unavailable.

The mock mode fallback implemented in model_loader.py ensures that the backend remains functional even when the model weights file is not available, which is useful for frontend development and testing purposes. The use of sigmoid rather than softmax in the inference stage correctly reflects the multi-label nature of the task, where a speaker can simultaneously express multiple emotions.

6.8 Comparison with the Research Questions

The research questions defined in Chapter 1 can now be addressed directly based on the experimental findings.

The first question asked how textual, acoustic, and visual features can be effectively combined for emotion recognition on CMU-MOSEI. The results show that bottleneck attention fusion, which routes cross-modal information through compact learnable tokens, is more effective than late fusion concatenation. The combination of frozen BERT text representations, Conv1D-projected COVAREP features, and Conv1D-projected OpenFace features, fused through two bottleneck layers, achieves the best performance among all configurations tested.

The second question asked whether an attention bottleneck mechanism improves multimodal fusion compared to a late fusion baseline. The answer is affirmative. The proposed bottleneck model outperforms the late fusion baseline by 4.42 percentage points in macro-F1 and achieves substantially better fear and surprise recognition. The improvement is consistent across all six emotion categories.

The third question asked how class imbalance affects the recognition of rare emotions. The results confirm that without active imbalance handling, models fail to recognize fear and surprise. With weighted loss, modality dropout, and threshold tuning, the proposed system achieves Fear F1 = 0.3451 and Surprise F1 = 0.3214, both exceeding the 0.30 benchmark reported in the literature [11].

The fourth question asked which evaluation metrics provide the most meaningful assessment of model performance. The results confirm that macro-F1 is the most informative metric for this task, as it treats all six emotion categories equally regardless of frequency. Binary accuracy and weighted accuracy are complementary but can mask poor minority class performance. Per-class F1 scores, especially for fear and surprise, provide the most granular assessment of a model's ability to handle the full emotional spectrum.

6.9 Summary

The discussion reveals five main findings. First, bottleneck attention fusion provides meaningful improvements over late fusion, confirming the value of structured cross-modal information exchange. Second, frozen BERT with averaged hidden layers is the optimal text strategy for this dataset size, balancing representation quality against overfitting risk. Third, the combination of weighted loss, modality dropout, and threshold tuning is necessary and effective for minority emotion recognition. Fourth, the primary performance ceiling on CMU-MOSEI with handcrafted features is the quality of audio and visual representations, not the fusion architecture. Fifth, the Semantic Enhancement Module does not generalize from neural encoder settings to handcrafted feature settings, which is an important finding for future work in this area.

Chapter 7

Conclusion

This project developed and evaluated a multimodal emotion recognition system that classifies six emotion categories from video input. The system processes three complementary streams — text transcripts, acoustic features, and facial visual features — and fuses them through an attention bottleneck mechanism. The work addressed the full research pipeline: literature analysis, dataset preparation, architecture design, training under class imbalance, quantitative evaluation, and integration into a functional prototype backend.

The proposed architecture combines three modality-specific encoders with a bottleneck fusion module. The text branch uses frozen BERT-base-uncased and averages the last four hidden layers to build word-level representations. This avoids overfitting that occurs when fine-tuning 110 million parameters on a dataset of 13 934 training samples. The audio and visual branches use Conv1D projections with positional encoding to bring COVAREP and OpenFace features into a shared 128-dimensional space. The fusion module uses 16 learnable bottleneck tokens as a compact communication channel. All cross-modal interaction passes through these tokens rather than through full pairwise attention between sequence positions, which keeps the complexity at $\mathcal{O}(T \cdot N_B)$ instead of $\mathcal{O}(T^2)$. Learnable modal weights control how much each stream contributes to the bottleneck state. Separate residual connections and LayerNorm instances guard each attention step. Auxiliary classification heads on each stream and modality dropout with probability 0.05 prevent the model from relying only on the strongest modality.

The experimental results confirm that bottleneck attention outperforms both implemented baselines. Baseline 1, which combines BERT with Conv1D audio encoding and a BiLSTM visual encoder using late fusion, reached Avg. F1 of 0.4412. Baseline 2, based on MulT-style pairwise cross-modal attention, improved this to 0.4631. The proposed bottleneck model achieved 0.4854 with the default decision threshold and 0.4942 after per-class threshold tuning on the validation set. This represents a 4.42 percentage point gain over Baseline 1 and shows that structured bottleneck communication adds value beyond both late fusion and unconstrained cross-attention.

Recognition of rare emotion categories improved substantially. Fear F1 rose from 0.1834 in Baseline 1 to 0.3122 at the default threshold and 0.3451 after tuning. Surprise F1 increased from 0.1902 to 0.3214 after tuning. These gains come from a combination of class-specific positive weights in the loss function, modality dropout regularisation, and per-class threshold selection on the validation set. Among published systems that use compact acoustic and visual descriptors, the proposed model surpasses MulT and approaches XMBT in Avg. F1.

The ablation study identified which design choices contributed most. Frozen BERT with averaged last four layers provided a stronger and more stable text representation than online fine-tuning. Auxiliary losses ensured that audio and visual branches retained task-relevant information throughout training. The Semantic Enhancement Module did not improve results in this setting, which shows that components from other architectures do not always transfer directly when the feature space differs.

The project also delivered a functional prototype. The FastAPI backend accepts raw video, extracts audio with ffmpeg, computes COVAREP features with opensmile, detects facial action units with py-feat, transcribes speech with faster-whisper, and returns emotion probabilities through a REST API. The system is currently a research prototype. Preprocessing, especially transcription and facial feature extraction, dominates inference time and limits real-time use.

The main limitations are the use of handcrafted acoustic and visual features, evaluation on a single English-language dataset, and the absence of cross-dataset testing. These constrain both the representational capacity of the model and the generalisability of the results.

Future work should replace COVAREP with pretrained speech encoders such as wav2vec2 or HuBERT, and OpenFace descriptors with a neural facial expression model. Additional directions include deeper bottleneck layers, larger token sets, focal loss for rare classes, and cross-dataset evaluation. On the engineering side, optimised preprocessing and GPU deployment would move the system closer to interactive use.

This project confirms that attention bottleneck fusion is an effective and computationally efficient strategy for multimodal emotion recognition. The developed system improves over both baselines, handles class imbalance more effectively, and provides a solid foundation for future work with stronger modality encoders.

Bibliography

[] Khan Mohammad, M. R. S., Kumar, S., Liu, Y., Song, R., and Behera, A. (2026).
Context-aware heterogeneous graph-driven multimodal representation learning for emotion recognition.
Submitted to ICLR 2026. OpenReview.

[2] He, P., Yu, J., Ge, C., Yu, Y., Xu, W., Wang, L., Liu, T., and Kan, Z. (2024).
Domain-separated bottleneck attention fusion framework for multimodal emotion recognition.
ACM Transactions on Multimedia Computing, Communications, and Applications, 37(4):Article 111.

[3] Jia, L., Tan, J., Qi, L., and Lin, M. (2024).
A more efficient inference model for multimodal emotion recognition.
In *Proceedings of Machine Learning Research*, volume 260.
Proceedings of ACML 2024.

[4] Kalateh, S., Estrada-Jimenez, L. A., Nikghadam-Hojjati, S., and Barata, J. (2024).
A systematic review on multimodal emotion recognition: Building blocks, current state, applications, and challenges.
IEEE Access, 12:1–30.

[5] Kawaguchi, K., Deng, Z., Ji, X., and Huang, J. (2023).
How does information bottleneck help deep learning?
In *Proceedings of the 40th International Conference on Machine Learning*, Proceedings of Machine Learning Research. PMLR.

[6] Kim, S., Kim, T.-S., and Lee, W. H. (2022).
Accelerating 3d convolutional neural network with channel bottleneck module for EEG-based emotion recognition.

Sensors, 22(18):6813.

[7] Koromilas, P. and Giannakopoulos, T. (2021).
Deep multimodal emotion recognition on human speech: A review.
Applied Sciences, 11(17):7962.

[8] Makhmudov, F., Kultimuratov, A., and Cho, Y.-I. (2024).
Enhancing multimodal emotion recognition through attention mechanisms in BERT and CNN architectures.
Applied Sciences, 14(10):4199.

[9] Mora Melanchthon, D. (2021).
Unimodal feature-level improvement on multimodal CMU-MOSEI dataset: Uncorrelated and convolved feature sets.
Procesamiento del Lenguaje Natural, (67):69–81.

[10] Nagrani, A., Yang, S., Arnab, A., Jansen, A., Schmid, C., and Sun, C. (2021).
Attention bottlenecks for multimodal fusion.
In *Advances in Neural Information Processing Systems*, volume 34, pages 14200–14213.

[11] Nguyen, D.-K., Lim, E., Kim, S.-H., Yang, H.-J., and Kim, S. (2025).
Enhanced emotion recognition through dynamic restrained adaptive loss and extended multimodal bottleneck transformer.
Applied Sciences, 15(5):2862.

[12] Pan, J., Fang, W., Zhang, Z., Chen, B., Zhang, Z., and Wang, S. (2023).
Multimodal emotion recognition based on facial expressions, speech, and EEG.
IEEE Open Journal of Engineering in Medicine and Biology, 3:392–401.

[13] Ramyasree, K., Rao, D. S., Rao, M. S., and Ahmad, S. M. K. M. A. (2025).
Emotion recognition through multimodal bottleneck transformers: The integration of speech and facial data to improve precision.
International Journal of Intelligent Engineering and Systems, 18(3):150–161.

[14] Sanjyal, A. (2026).
Multimodal emotion recognition via causal-diffusion bridge (Affect-Diff).

[15] Saxe, A. M., Bansal, Y., Dapello, J., Advani, M., Kolchinsky, A., Tracey, B. D., and Cox, D. D. (2018).
On the information bottleneck theory of deep learning.

In *International Conference on Learning Representations*.

[16] Srinivas, A., Lin, T.-Y., Parmar, N., Shlens, J., Abbeel, P., and Vaswani, A. (2021).
Bottleneck transformers for visual recognition.
In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 16519–16529.

[17] Tsai, Y.-H. H., Bai, S., Liang, P. P., Kolter, J. Z., Morency, L.-P., and Salakhutdinov, R. (2019).
Multimodal transformer for unaligned multimodal language sequences.
In *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics*, pages 6558–6569.

[18] Xia, X., Zhao, Y., and Jiang, D. (2022).
Multimodal interaction enhanced representation learning for video emotion recognition.
Frontiers in Neuroscience, 16:1086380.

[19] Yuan, C., Jin, Y., Wu, Z., Wei, F., Wang, Y., Chen, L., and Wang, X. (2023).
Learning bottleneck transformer for event image-voxel feature fusion based classification.
arXiv preprint arXiv:2308.11937.

[20] Zadeh, A., Liang, P. P., Vanbriesen, J., Poria, S., Cambria, E., Chen, M., and Morency, L.-P. (2018).
Multimodal language analysis in the wild: CMU-MOSEI dataset and interpretable dynamic fusion graph.
In *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics*, pages 2236–2246.

Appendix A

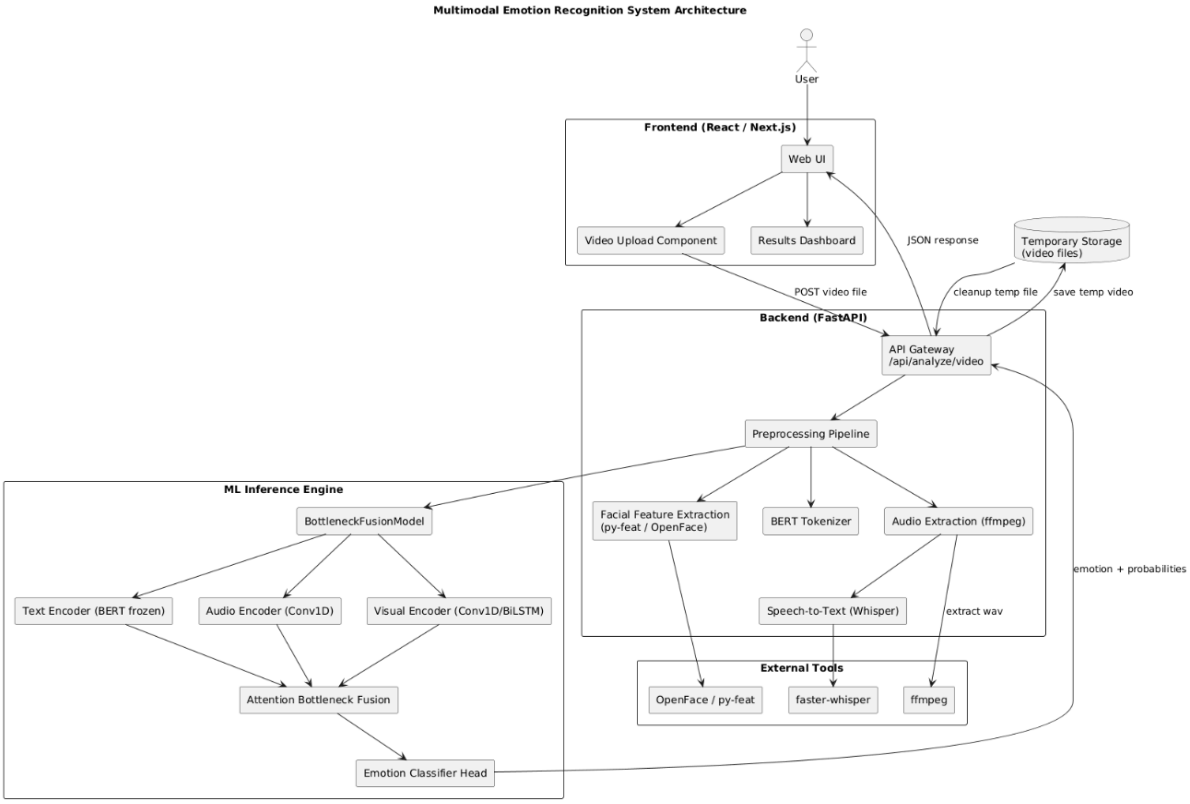

Figure 1: System Architecture

Appendix B

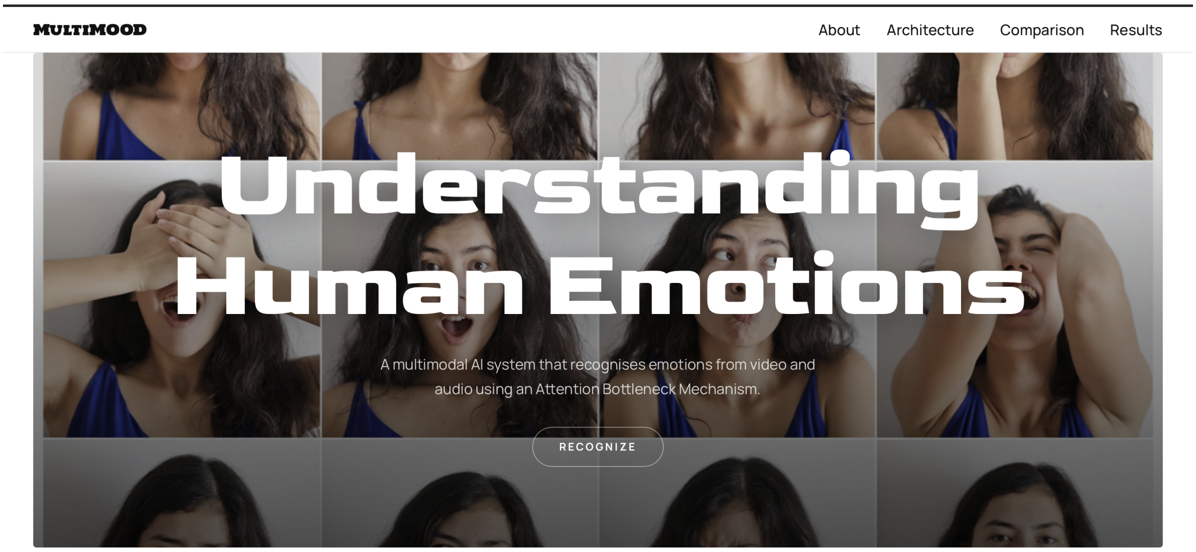

Figure 2: Frontend interface of the emotion recognition system

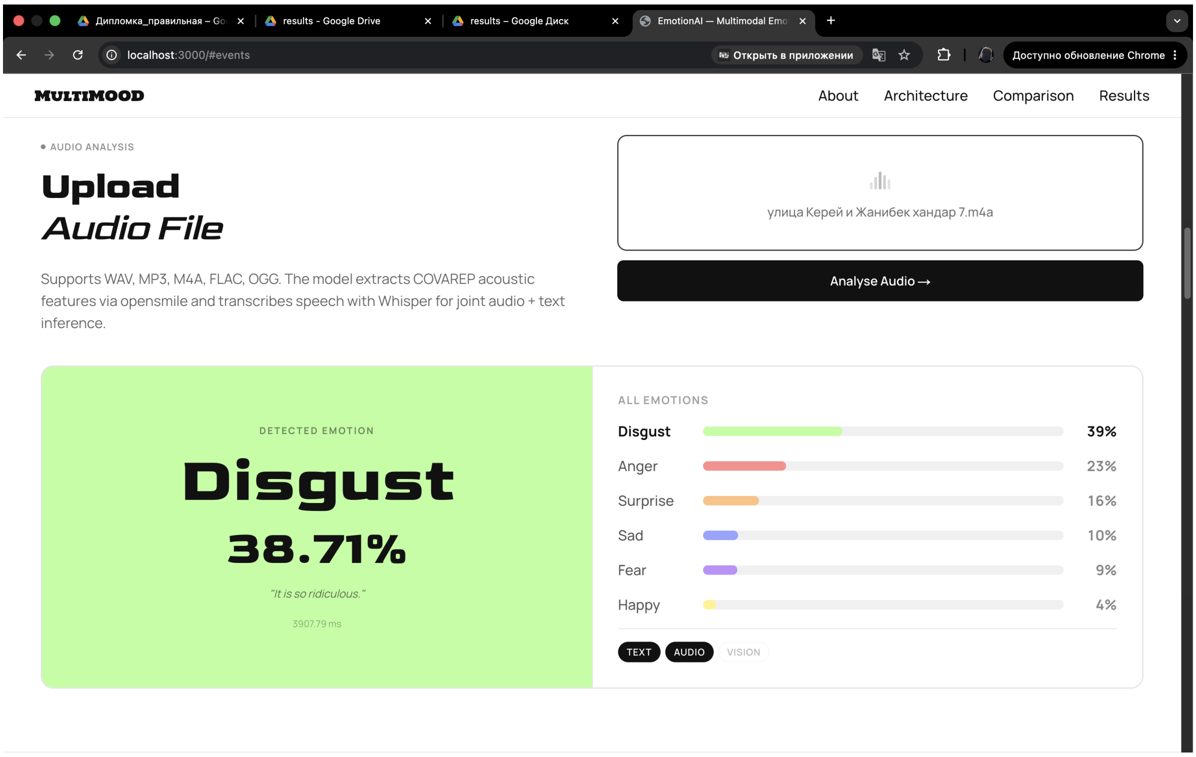

Figure 3: Audio modality prediction results

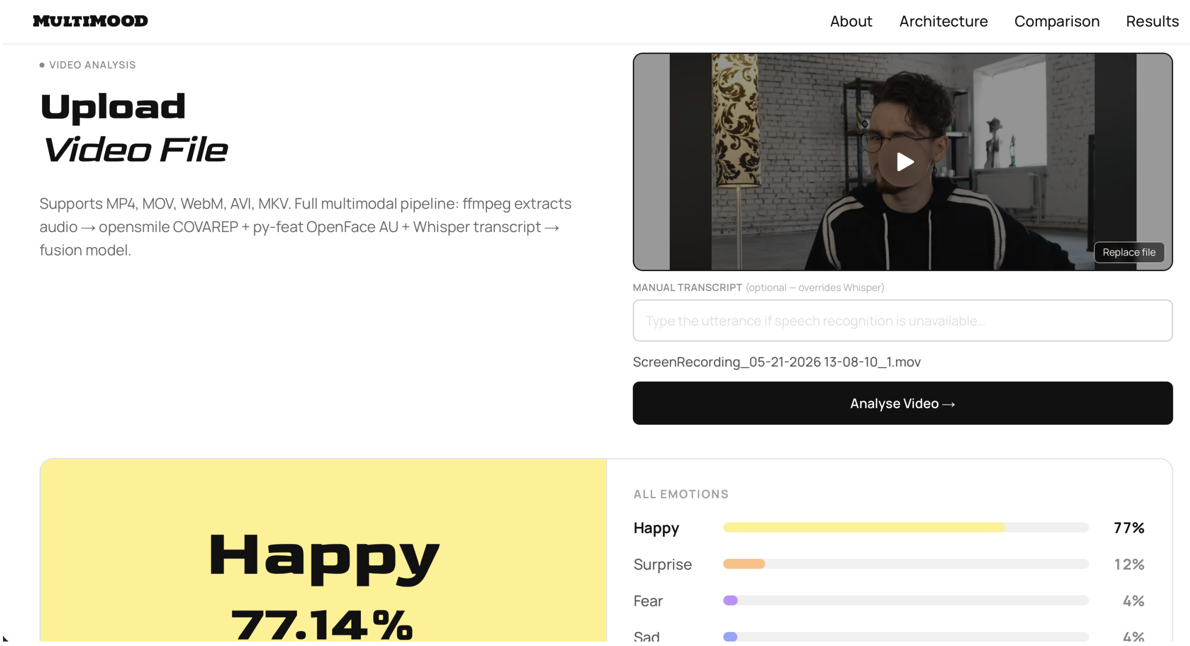

Figure 4: Video modality prediction results

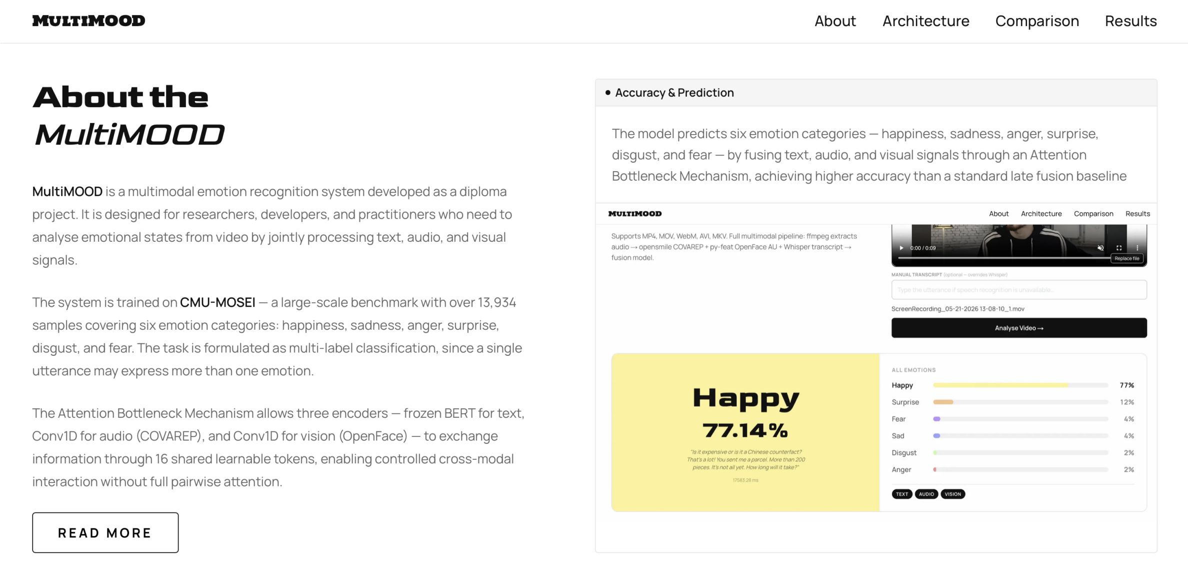

Figure 5: About System

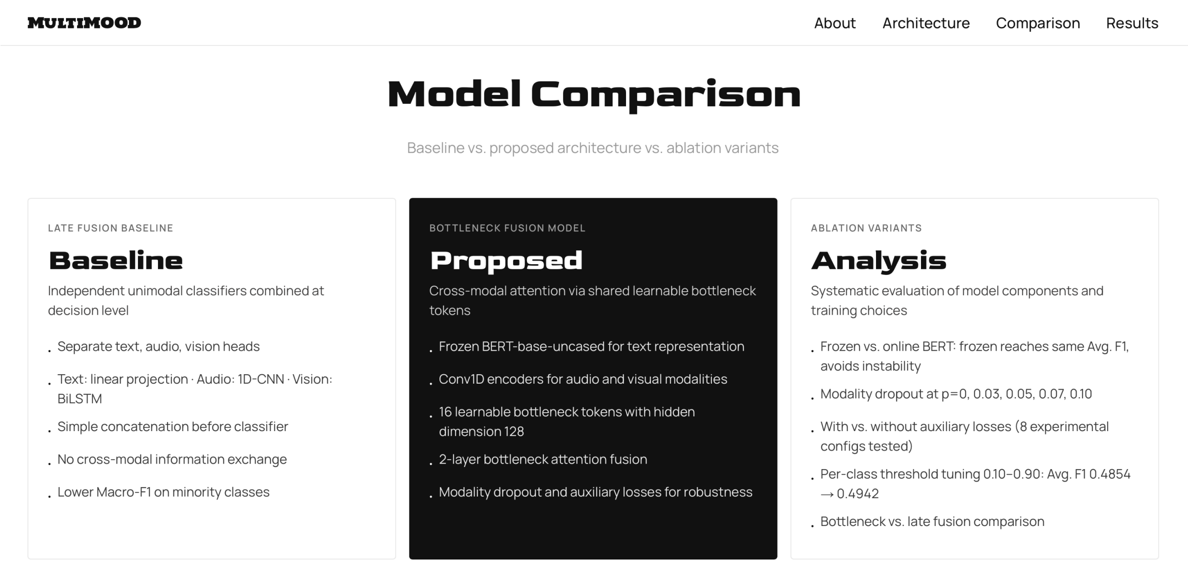

Figure 6: Models comparison

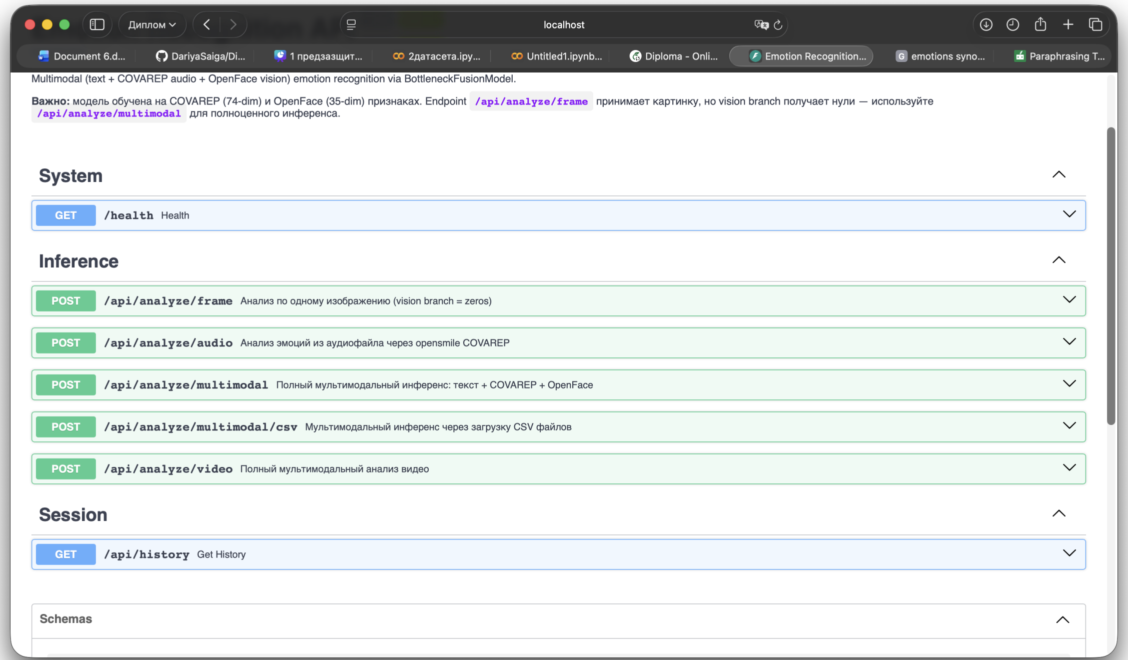

Figure 7: Backend API endpoint configuration

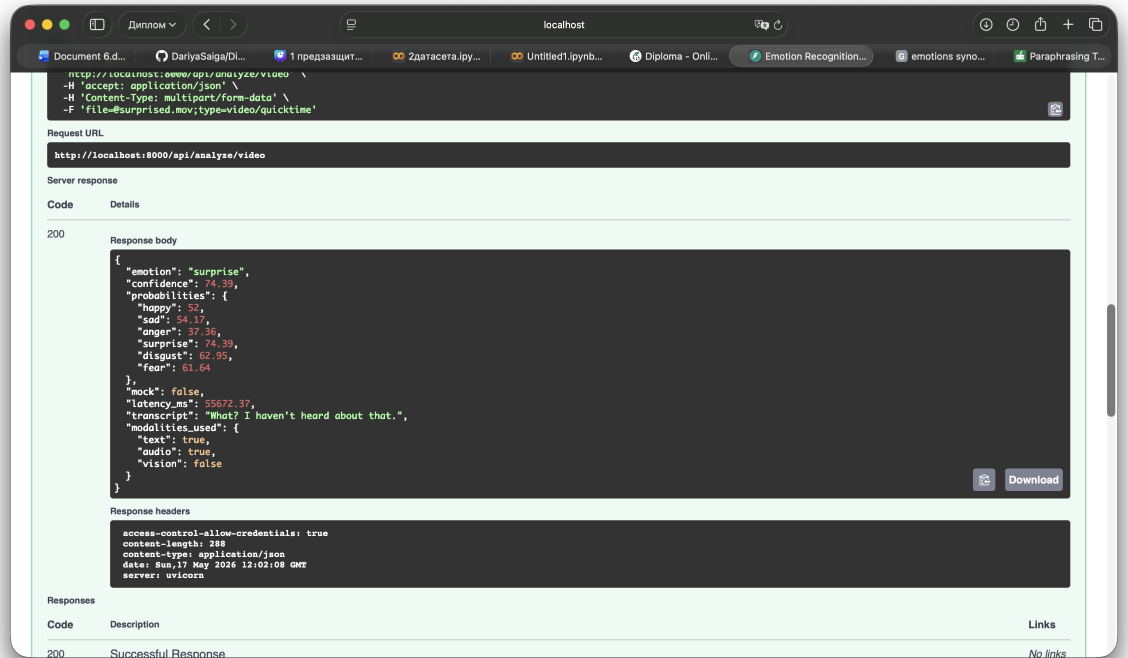

Figure 8: Backend processing result output

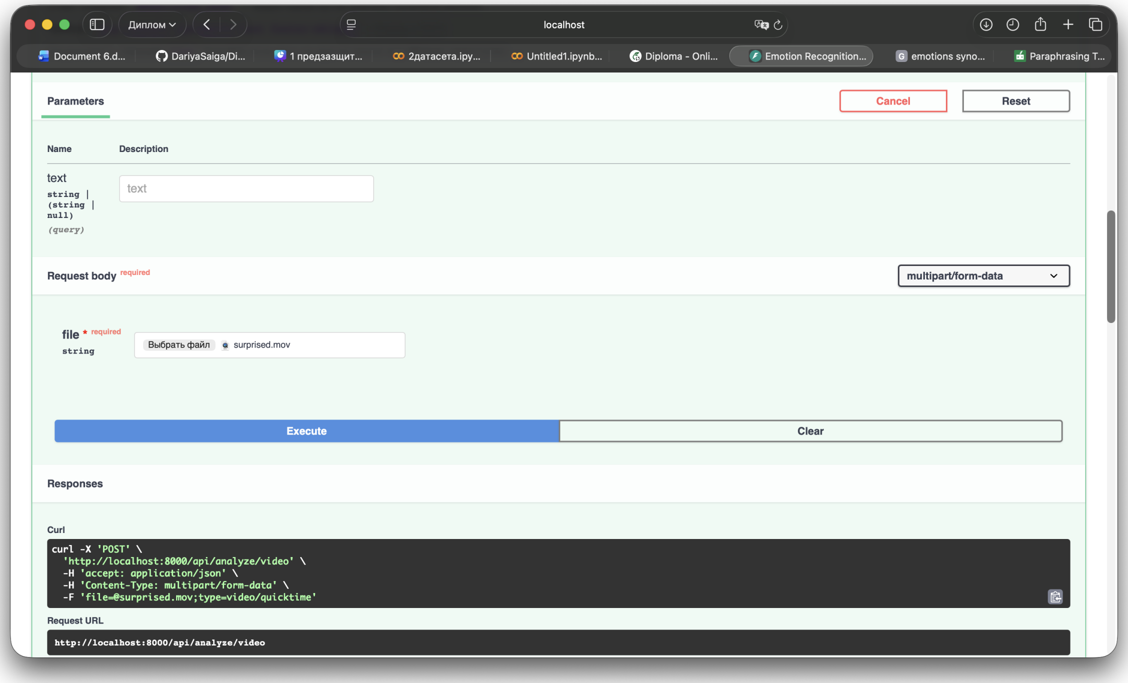

Figure 9: Backend server interface

Appendix C

```python
class BottleneckAttentionFusion(nn.Module):
    def __init__(self, d_model, num_heads=4, num_bottleneck=16, dropout=0.1):
        self.attn_bt = nn.MultiheadAttention(d_model, num_heads, dropout=dropout, batch_first=True)
        self.attn_ba = nn.MultiheadAttention(d_model, num_heads, dropout=dropout, batch_first=True)
        self.attn_bv = nn.MultiheadAttention(d_model, num_heads, dropout=dropout, batch_first=True)

        self.ffn = nn.Sequential(
            nn.Linear(d_model, d_model * 4),
            nn.GELU(),
            nn.Dropout(dropout),
            nn.Linear(d_model * 4, d_model),
        )

        self.norm_t   = nn.LayerNorm(d_model)
        self.norm_a   = nn.LayerNorm(d_model)
        self.norm_v   = nn.LayerNorm(d_model)
        self.norm_b   = nn.LayerNorm(d_model)
        self.norm_ffn = nn.LayerNorm(d_model)
        self.drop     = nn.Dropout(dropout)

    def forward(self, t, a, v, bn=None):
        B  = t.size(0)
        if bn is None:
            bn = self.bottleneck.expand(B, -1, -1)

        bt, _ = self.attn_bt(bn, t, t)
        ba, _ = self.attn_ba(bn, a, a)
        bv, _ = self.attn_bv(bn, v, v)
        bn = self.norm_b(bn + self.drop(bt + ba + bv) / 3.0)

        t2, _ = self.attn_t(t, bn, bn)
        a2, _ = self.attn_a(a, bn, bn)
        v2, _ = self.attn_v(v, bn, bn)

        t = self.norm_t(t + self.drop(t2))
        a = self.norm_a(a + self.drop(a2))
        v = self.norm_v(v + self.drop(v2))

        bn = self.norm_ffn(bn + self.drop(self.ffn(bn)))
        return bn, t, a, v
```

Figure 10: Bottleneck Attention Fusion mechanism used in the multimodal model

Bottleneck Attention Layer

```python
class BottleneckLayer(nn.Module):
    def __init__(self):
        super().__init__()
        # self-attention внутри каждой модальности
        self.self_attn_t = nn.MultiheadAttention(HIDDEN_DIM, N_HEADS, DROPOUT, batch_first=True)
        self.self_attn_a = nn.MultiheadAttention(HIDDEN_DIM, N_HEADS, DROPOUT, batch_first=True)
        self.self_attn_v = nn.MultiheadAttention(HIDDEN_DIM, N_HEADS, DROPOUT, batch_first=True)

        # cross-attention: каждая модальность → bottleneck tokens (сжатие)
        self.cross_t2b = nn.MultiheadAttention(HIDDEN_DIM, N_HEADS, DROPOUT, batch_first=True)
        self.cross_a2b = nn.MultiheadAttention(HIDDEN_DIM, N_HEADS, DROPOUT, batch_first=True)
        self.cross_v2b = nn.MultiheadAttention(HIDDEN_DIM, N_HEADS, DROPOUT, batch_first=True)

        # cross-attention: bottleneck tokens → каждая модальность (расширение)
        self.cross_b2t = nn.MultiheadAttention(HIDDEN_DIM, N_HEADS, DROPOUT, batch_first=True)
        self.cross_b2a = nn.MultiheadAttention(HIDDEN_DIM, N_HEADS, DROPOUT, batch_first=True)
        self.cross_b2v = nn.MultiheadAttention(HIDDEN_DIM, N_HEADS, DROPOUT, batch_first=True)

        # FFN блоки после attention для каждой модальности
        self.ffn_t = self._make_ffn()
        self.ffn_a = self._make_ffn()
        self.ffn_v = self._make_ffn()

        # LayerNorm
        self.norm_t = nn.LayerNorm(HIDDEN_DIM)
        self.norm_a = nn.LayerNorm(HIDDEN_DIM)
        self.norm_v = nn.LayerNorm(HIDDEN_DIM)

    def _make_ffn(self):
        return nn.Sequential(
            nn.Linear(HIDDEN_DIM, HIDDEN_DIM * 4),
            nn.GELU(),
            nn.Dropout(DROPOUT),
            nn.Linear(HIDDEN_DIM * 4, HIDDEN_DIM),
            nn.Dropout(DROPOUT),
        )
```

Figure 11: BottleneckLayer implementation in models.py

Dataset Preprocessing

```python
def get_dataloaders():

    # ── Проблема 8: воспроизводимость ──────────────────────────────────────
    # Статья: DBA (He et al., 2024) — фиксированные гиперпараметры и воспроизводимость.
    torch.manual_seed(42)
    np.random.seed(42)

    print("Загружаем датасет...")
    with open(PKL_PATH, "rb") as f:
        data = pickle.load(f, encoding="latin1")

    loaders = {}
    for split in ["train", "valid", "test"]:
        print(f"\n[{split}]")
        ds = MOSEIDataset(data[split])
        loaders[split] = DataLoader(
            ds,
            batch_size=BATCH_SIZE,
            shuffle=(split == "train"),
            num_workers=NUM_WORKERS,
            pin_memory=True,
        )

    # pos_weight считаем только по train
    train_labels = loaders["train"].dataset.labels.numpy()
    pos_weight   = get_pos_weight(train_labels)

    print("\n── pos_weight (BCEWithLogitsLoss) ──")
    emotions = ["happy", "sad", "anger", "surprise", "disgust", "fear"]
    for emo, pw in zip(emotions, pos_weight):
        print(f"  {emo:<10}: {pw:.2f}")

    return loaders, pos_weight
```

Figure 12: Dataset loading and preprocessing pipeline

Feature Extractor

```python

detector = Detector(
    face_model="retinaface",
    landmark_model="mobilefacenet",
    au_model="xgb",
    facepose_model="img2pose",
    emotion_model="resmasknet",
    device="cpu",
)

# ── 2. Читаем 1 кадр из видео ──────────────────────────────────────────
cap = cv2.VideoCapture(video_path)
total_frames = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))

if total_frames <= 0:
    cap.release()
    _err("Не удалось прочитать видео (total_frames=0)")
    sys.exit(1)

frame_indices = np.linspace(0, total_frames - 1, 15, dtype=int)
tmp_dir = Path(tempfile.gettempdir())

for i, idx in enumerate(frame_indices):
    cap.set(cv2.CAP_PROP_POS_FRAMES, int(idx))
    ok, frame = cap.read()
    if not ok:
        continue
    # Минимальный размер для img2pose - 400px
    frame = cv2.resize(frame, (400, 400))
    frame_path = tmp_dir / f"feat_frame_{uuid.uuid4().hex}_{i}.jpg"
    cv2.imwrite(str(frame_path), frame)
    frame_paths.append(str(frame_path))

cap.release()
```

Figure 13: Visual feature extraction using py-feat

Training Loop

```python
LR_MAIN      = 1e-4   # lr основной сети — DBA (He et al., 2024): lr=1e-4
LR_BERT      = 5e-6   # lr BERT — XMBT (Nguyen et al., 2025): text lr = lr_main / 10
EPOCHS       = 50     # DBA: max 80, XMBT: 30 — берём 50 как компромисс
PATIENCE     = 5      # XMBT: early stopping patience = 6 эпох
GRAD_CLIP    = 1.0    # MulT (Tsai et al., 2019): gradient clip = 1.0

SAVE_PATH    = "/content/drive/MyDrive/Дипломка_правильная/results/best_model.pt"

# ============================================================================

EMOTIONS = ["happy", "sad", "anger", "surprise", "disgust", "fear"]


def get_device():
    if torch.backends.mps.is_available():
        return torch.device("mps")
    if torch.cuda.is_available():
        return torch.device("cuda")
    return torch.device("cpu")


def train_epoch(model, loader, optimizer, criterion, device):
    """Один проход по train — возвращает средний loss."""
    model.train()
    total_loss = 0.0

    for batch in loader:
        input_ids      = batch["input_ids"].to(device)
        attention_mask = batch["attention_mask"].to(device)
        audio          = batch["audio"].to(device)
        vision         = batch["vision"].to(device)
        audio_mask     = batch["audio_mask"].to(device)
        vision_mask    = batch["vision_mask"].to(device)
```

Figure 14: Training loop implementation

Train and Evaluate Functions

```python
@torch.no_grad()
def evaluate(model, loader, criterion, device):
    """
    Валидация — считаем loss, Avg. F1, Avg. WA, per-class F1.
    Статья: MER-SEM-MBT, XMBT — главные метрики для CMU-MOSEI:
    Average F1 (macro) и Average Weighted Accuracy (Avg. WA)
    """
    model.eval()
    total_loss = 0.0
    all_preds  = []
    all_labels = []

    for batch in loader:
        input_ids      = batch["input_ids"].to(device)
        attention_mask = batch["attention_mask"].to(device)
        audio          = batch["audio"].to(device)
        vision         = batch["vision"].to(device)
        audio_mask     = batch["audio_mask"].to(device)
        vision_mask    = batch["vision_mask"].to(device)
        labels         = batch["labels"].to(device)

        logits_fuse, _, _, _ = model(input_ids, attention_mask, audio, vision, audio_mask, vision_mask)
        loss = criterion(logits_fuse, labels)
        total_loss += loss.item()

        # sigmoid → бинаризация порогом 0.5
        preds = (torch.sigmoid(logits_fuse) > 0.5).cpu().numpy()
        all_preds.append(preds)
        all_labels.append(labels.cpu().numpy())

    all_preds  = np.vstack(all_preds)   # [N, 6]
    all_labels = np.vstack(all_labels)  # [N, 6]

    # —— Avg. F1 (macro) — главная метрика ————————————————————————————————
    # Статья: MER-SEM-MBT — "Avg. F1 is used as the main evaluation indicator"
    avg_f1 = f1_score(all_labels, all_preds, average="macro", zero_division=0)
```

Figure 15: train_epoch and evaluate functions

Video Feature Extraction

```python
logging.basicConfig(level=logging.WARNING, stream=sys.stderr)
for _lg in ("feat", "feat.detector", "py_feat", "huggingface_hub", "transformers"):
    logging.getLogger(_lg).setLevel(logging.ERROR)

VISION_SEQ_LEN  = 60
VISION_FEAT_DIM = 35


def _extract(detector, video_path: str) -> np.ndarray:
    frame_paths: list[str] = []
    tmp_dir = Path(tempfile.gettempdir())
    try:
        cap = cv2.VideoCapture(video_path)
        total = int(cap.get(cv2.CAP_PROP_FRAME_COUNT))
        if total <= 0:
            raise ValueError(f"Не удалось прочитать видео: {video_path}")

        for i, idx in enumerate(np.linspace(0, total - 1, 8, dtype=int)):
            cap.set(cv2.CAP_PROP_POS_FRAMES, int(idx))
            ok, frame = cap.read()
            if not ok:
                continue
            frame = cv2.resize(frame, (400, 400))
            p = tmp_dir / f"feat_{uuid.uuid4().hex}_{i}.jpg"
            cv2.imwrite(str(p), frame)
            frame_paths.append(str(p))
        cap.release()

        if not frame_paths:
            raise ValueError("Не удалось извлечь кадры")

        result = detector.detect_image(frame_paths)
```

Figure 16: Video preprocessing and feature extraction pipeline